MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE ADH

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic JK flip-flop module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.52 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein

3.5.1 Key code. The module key code shall be ADH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 75 grams

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5 7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/123-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3 7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4 3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665 Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4 4 Test procedures The module shall be tested as specified in table I

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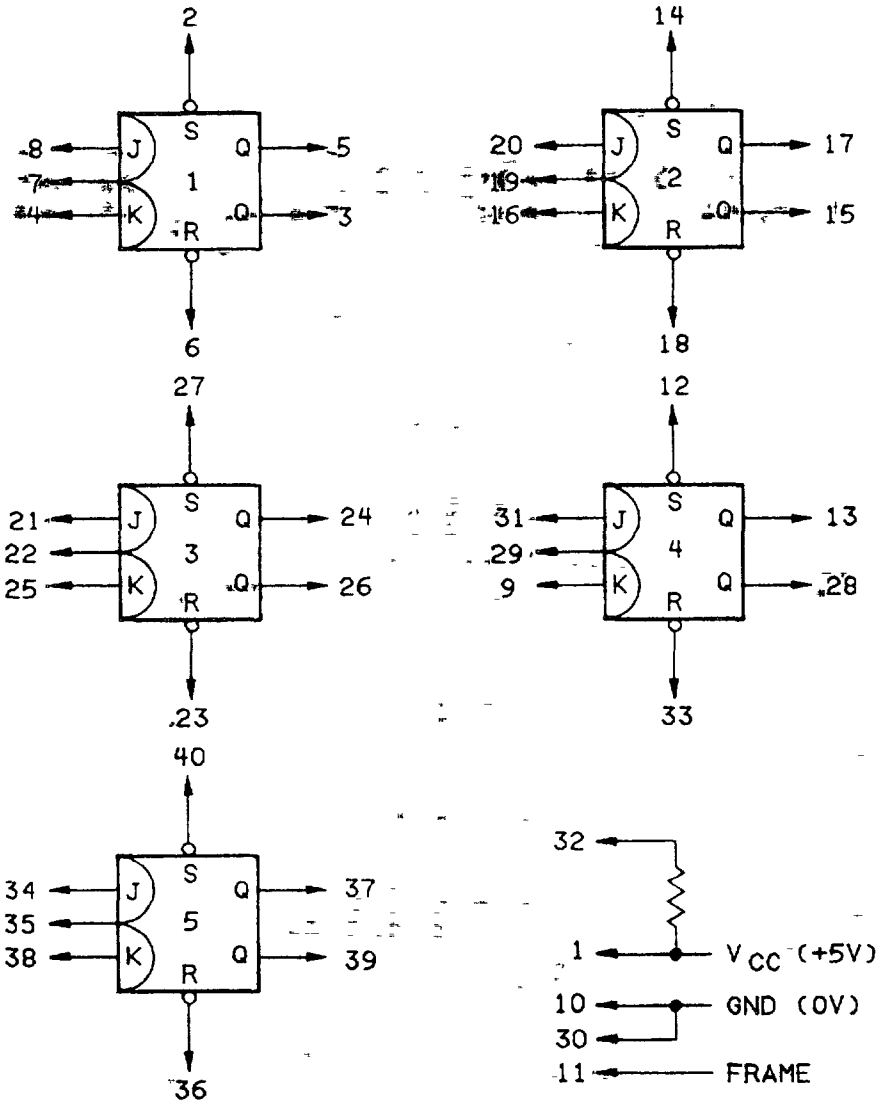


FIGURE 1. ADH module function diagram, J K flip-flop

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TABLE I Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A25	Zener applied potential, cond A	III	S	3 8	5 5	3 5	5 5	3.8	5 5	3.5	5.5	V
B1-B5	Zener applied potential, cond B	III	S	3.8	5 5	3.5	5 5	3 8	5 5	3 5	5 5	V
C1-C10	V_{OUT} (low loaded)	IV	S	0 0	1 1	0 0	1 1	0 0	1 I	0 0	1 1	V
D	Isolation	V	S	10	∞	10	∞	10	∞	10	∞	M
E	Resistance	V	X	4	5	4	5	4	5	4	5	k
F1	Continuity	V	X	0	1	0	1	0	1	0	1	
F2	Continuity	V	S	0	1	0	1	0	1	0	1	
G1-G5	Input current, reverse clock	VI	X	0	4	0	8	0	6	0	10	µA
H1-H10	Input reverse current, synchronous	VII	X	0	2	0	4	0	3	0	5	µA
J1-J10	Input reverse current, asynchronous	VII	X	0	4	0	8	0	6	0	10	µA
K1-K5	Input forward current, J	VIII	X	0 20	1 07	0 20	1 07	0 20	1 07	0 20	1.07	mA
K6-K10	Input forward current, K	VIII	X	0.20	1 07	0 20	1 07	0 20	1 07	0.20	1.07	mA
L1-L5	Input forward current, R	VIII	X	0.2	3.2	0 2	3 2	0.2	3 2	0 2	3 2	mA
L6-L10	Input forward current, S	VIII	X	0 2	3 2	0 2	3 2	0 2	3 2	0 2	3 2	mA
M1-M10	Input forward current, clock	VIII	X	0 2	3.2	0 2	3 2	0.2	3 2	0 2	3.2	mA

See footnotes at end of table

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1-N5	Output voltage, high level (J input high level threshold)	IX	X	3 8	5 5	3.5	5 5	3.8	5 5	3 5	5.5	V
P1-P5	Output voltage, high level (K input high level threshold)	IX	X	3 8	5 5	3 5	5 5	3.8	5.5	3 5	5.5	V
Q1-Q5	Output voltage, high level (S input low level threshold)	IX	X	3 8	5 5	3.5	5.5	3.8	5 5	3 5	5.5	V
R1-R5	Output voltage, high level (R input low level threshold)	IX	X	3 8	5 5	3 5	5.5	3 8	5.5	3 5	5.5	V
S1-S5	Output voltage, low level (S input high level threshold)	IX	X	0 00	0 40	0 00	0.45	0 00	0 40	0 00	0.45	V
T1-T5	Output voltage, low level (R input high level threshold)	IX	X	0 00	0.40	0.00	0 45	0.00	0.40	0 00	0 45	V
U1-U10	Output short circuit current	X	X	0 9	3 0	0.9	3.0	0.9	3 0	0.9	3.0	mA
V1-V2	Power supply current	X	S	0 0	82 5	0.0	82.5	0 0	82.5	0.0	82.5	mA
W1-W5	Output voltage, high level (J input low level threshold)	XI	X	3 8	5 5	3.5	5.5	3.8	5.5	3 5	5.5	V
X1-X5	Output voltage, high level (K input low level threshold)	XI	X	3 8	5 5	3.5	5.5	3.8	5.5	3 5	5 5	V
Y1-Y5	Operation, J-K	XII	X	16	24	16	24	16	24	16	24	µs

See footnotes at end of table

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TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Z1-Z5	Parallel output 1	XIII	Q	0 00	0 40	0.00	0.45	0 00	0 40	0 00	0.45	V
Z6-Z10	Parallel output 2	XIII	Q	0 00	0 40	0 00	0 45	0 00	0 40	0 00	0 45	V
Z11-Z15	Parallel output 3	XIII	Q	0 00	0 40	0 00	0 45	0.00	0 40	0 00	0 45	V
Z16-Z20	Parallel output 4	XIII	Q	0 00	0 40	0 00	0 45	0 00	0.40	0 00	0 45	V
AA1-AA10	Transition time (t_{TLH})	XIV	S	55	170	55	170	55	170	55	170	ns
AA11-AA20	Transition time (t_{THL})	XIV	S	6	24	6	24	6	24	6	24	ns
BB1-BB10	Propagation delay time, t_{PHL} synchronous	XV	S	20	80	20	80	20	80	20	80	ns
BB11-BB20	Propagation delay time, t_{PLH} synchronous	XV	S	30	90	30	90	30	90	30	90	ns
CC1-CC10	Output voltage, high level	XVI	S	3 8	5.5	3 5	5 5	3 8	5.5	3 5	5 5	V
CC11-CC20	Output voltage, low level	XVI	S	0.00	0 40	0 00	0 45	0 00	0 40	0 00	0 45	V
DD1-DD5	Transient immunity (3V), clock	XVII	S	0.00	0 40	0 00	0.45	0 00	0 40	0 00	0 45	V
DD6-DD10	Transient immunity (I 5V), clock	XVII	S	0 00	0 40	0 00	0 45	0 00	0 40	0 00	0 45	V
EE1-EE5	Clock threshold, high level slow transition times	XVIII	S	4	6	4	6	4	6	4	6	ms
FF1-FF10	Clock threshold, low level	XVIII	S	0 00	0 40	0 00	0 45	0 00	0 40	0 00	0 45	V
GG1-GG10	Transient immunity, J and K	XIX	Q	0.00	0.40	0 00	0 45	0 00	0 40	0 00	0 45	V

See footnotes at end of table.

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
HH1-HH5	Minimum pulse sensitivity, J-K	XIX	Q	1 6	2 4	1.6	2.4	1 6	2 4	1.6	2.5	µs
JJ	Power supply transient rise time	XX	S	36	148	36	148	36	148	36	148	µs
KK1-KK10	Output crosstalk, low level and high level	XXI	Q	0.0	0 3	0 0	0 3	0 0	0 3	0 0	0 3	V
LL1-LL25	Input capacitance	XXII	Q	2	16	2	16	2	16	2	16	pF
MM1-MM10	Propagation delay time, t_{PHL}, R and S	XXIII	Q	15	80	15	80	15	80	15	80	ns
MM11-MM20	Propagation delay time, t_{PLH}, R and S	XXIII	Q	15	95	15	95	15	95	15	95	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through MM shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through MM in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E, F1, G through KK, and MM shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5 Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E, F1, G through KK, and MM shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E, F1 and G through MM shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/123 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658629. Part number M28787/123-1 supersedes part numbers 2658629 and M28787/123.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Power dissipation	454 mW maximum
Input voltage	
Low level	0.0 V minimum, 1.2 V maximum
High level	1.8 V minimum, 5.0 V maximum
Speed	DC to 0.6 MHz

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6.5 Module function. The module consists of five independent JK flip-flops with set and reset inputs and Q and $\overline{Q}$ outputs. A pull up resistor to V_{CC} is also available.

6.5.1 Input to output relationship. Each flip-flop has the input to output relationship for positive logic shown in table II.

6.5.2 Paralleling of outputs (wired OR). Paralleling of outputs is permissible with a reduction in low level fan out calculated from the following expression.

Reduced low level fan out = maximum fan out -1.9 (n-1)

Where maximum fan out = 10 S.L.L (standard logic load) low level and n = number of outputs connected in parallel.

The increase in high level fan out is calculated from the following expression.

Increased high level fan out = maximum fan out (n)

Where maximum fan out = 12 S.L.L. high level and n = number of outputs connected in parallel.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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TABLE II. Input to output relationship. 1/-8/

Asynchronous inputs		Synchronous inputs		Clock input	Outputs	
Set	Reset	J	K		Q	$\bar{Q}$
L	L	X	X	X	H	H
H	L	X	X	X	L	H
L	H	X	X	X	H	L
H	H	L	H	H L	L	H
H	H	H	L	H L	H	L
H	H	L	L	H L	NC	NC
H	H	H	H	H L	CS	CS

1/ H = High level.

2/ L = Low level.

3/ H L = High to low transition of clock pulse.

4/ X = Don't care.

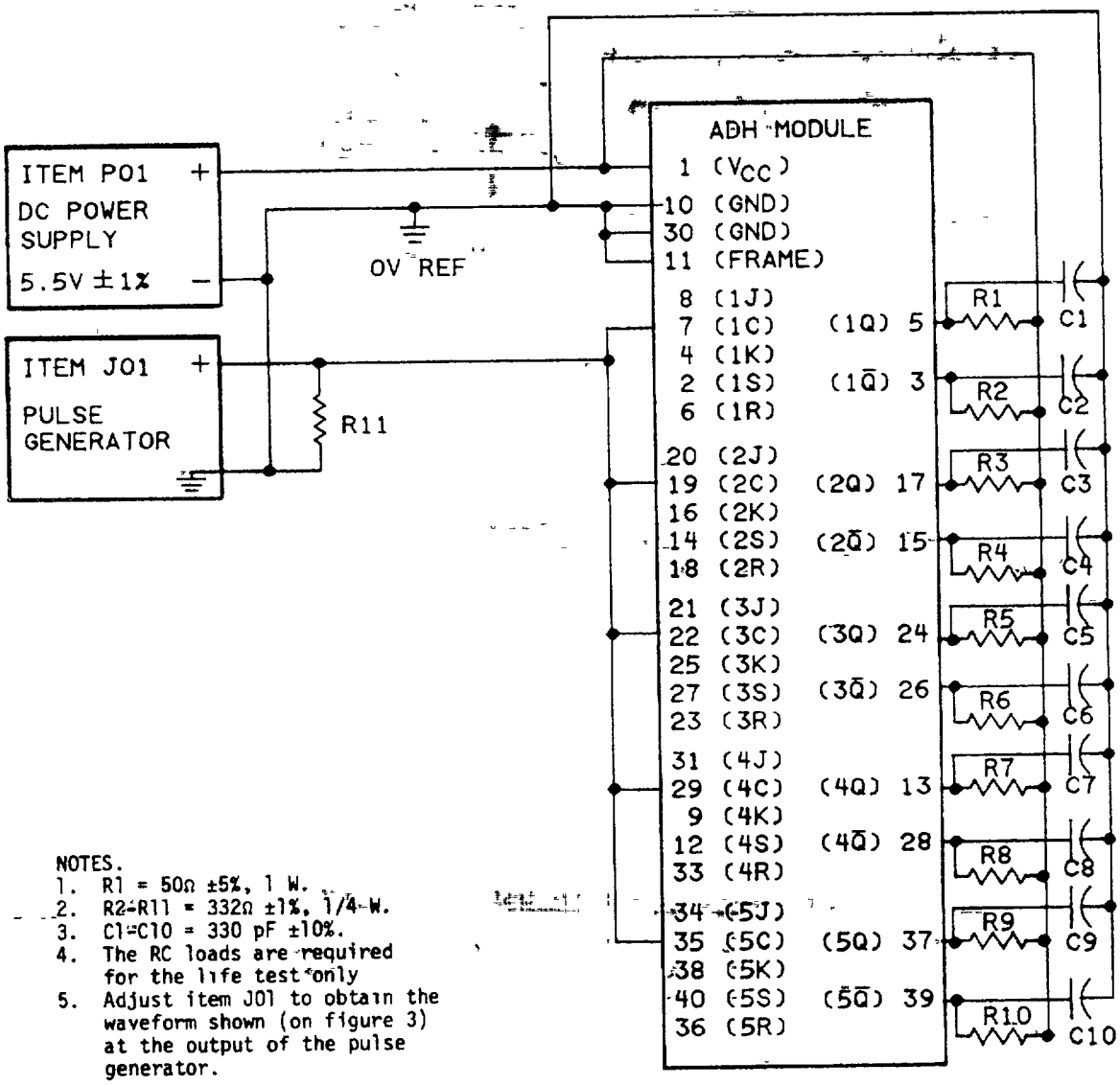
5/ NC = No change from previous state

6/ CS = Change from previous state.

7/ Set and reset are independent of clock.

8/ J and K are synchronous with high to low transition of clock

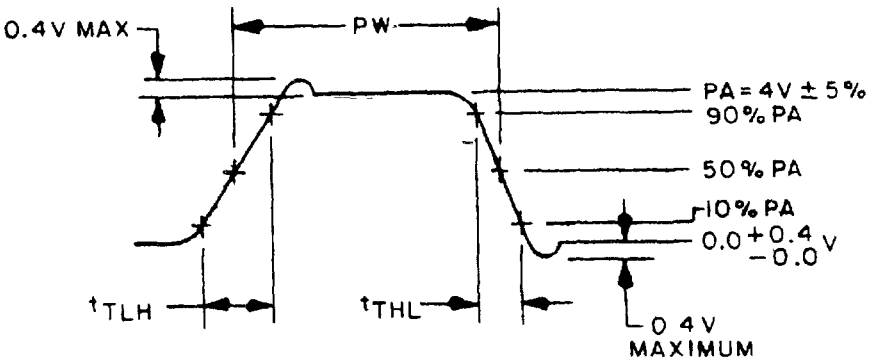
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NOTES.
1. R1 = 50Ω ±5%, 1 W.
2. R2-R11 = 332Ω ±1%, 1/4 W.
3. C1-C10 = 330 pF ±10%.
4. The RC loads are required for the life test only
5. Adjust item JO1 to obtain the waveform shown (on figure 3) at the output of the pulse generator.

FIGURE 2. Operating temperature and life test circuit

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NOTE Input pulse parameters
$t_{TLH} = t_{TTHL} = 50 \pm 5$ ns.
PW = 0.5 µs ±10%.
PRR = 1.0 MHz ±10%

FIGURE 3. Operating temperature and life test circuit waveform

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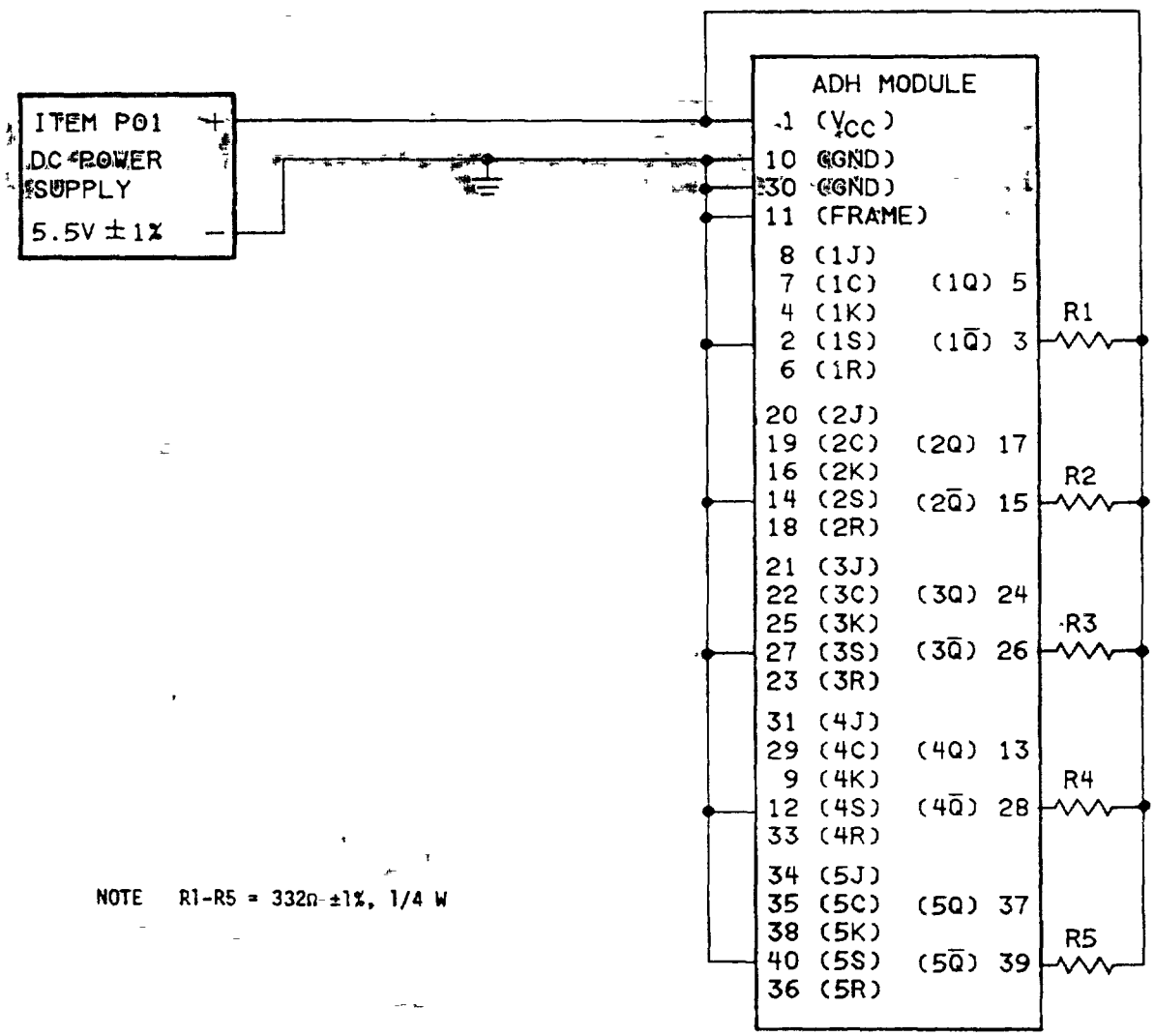


FIGURE 4 Component temperature test circuit.

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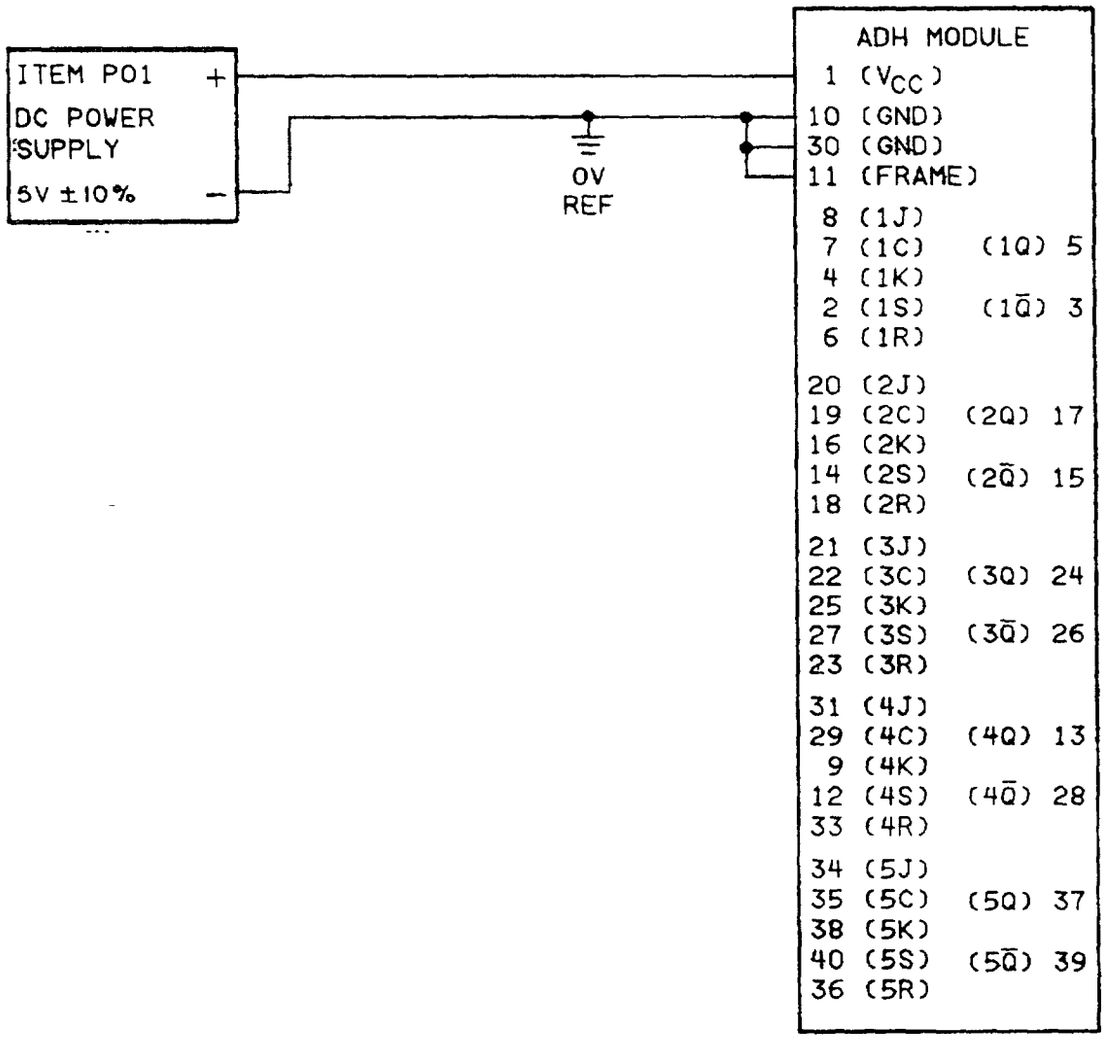


FIGURE 5 Vibration test circuit.

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TABLE III. Zener applied potential (cond A and cond B) test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
A1	1Q Zener applied potential, cond A	Open	5	2	1	8	3.8	5.5	V	6,7
A2	1Q					7				
A3	1Q					4				
A4	1Q					6				
A5	$\overline{1Q}$		3	6		2				
A6	2Q		17	14		20				
A7	2Q					19				
A8	2Q					16				
A9	2Q					18				
A10	$\overline{2Q}$		15	18		14				
A11	3Q		24	27		21				
A12	3Q		24	27		22				
A13	3Q		24	27		25				

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TABLE III. Zener applied potential (cond A and cond B) test - Continued.

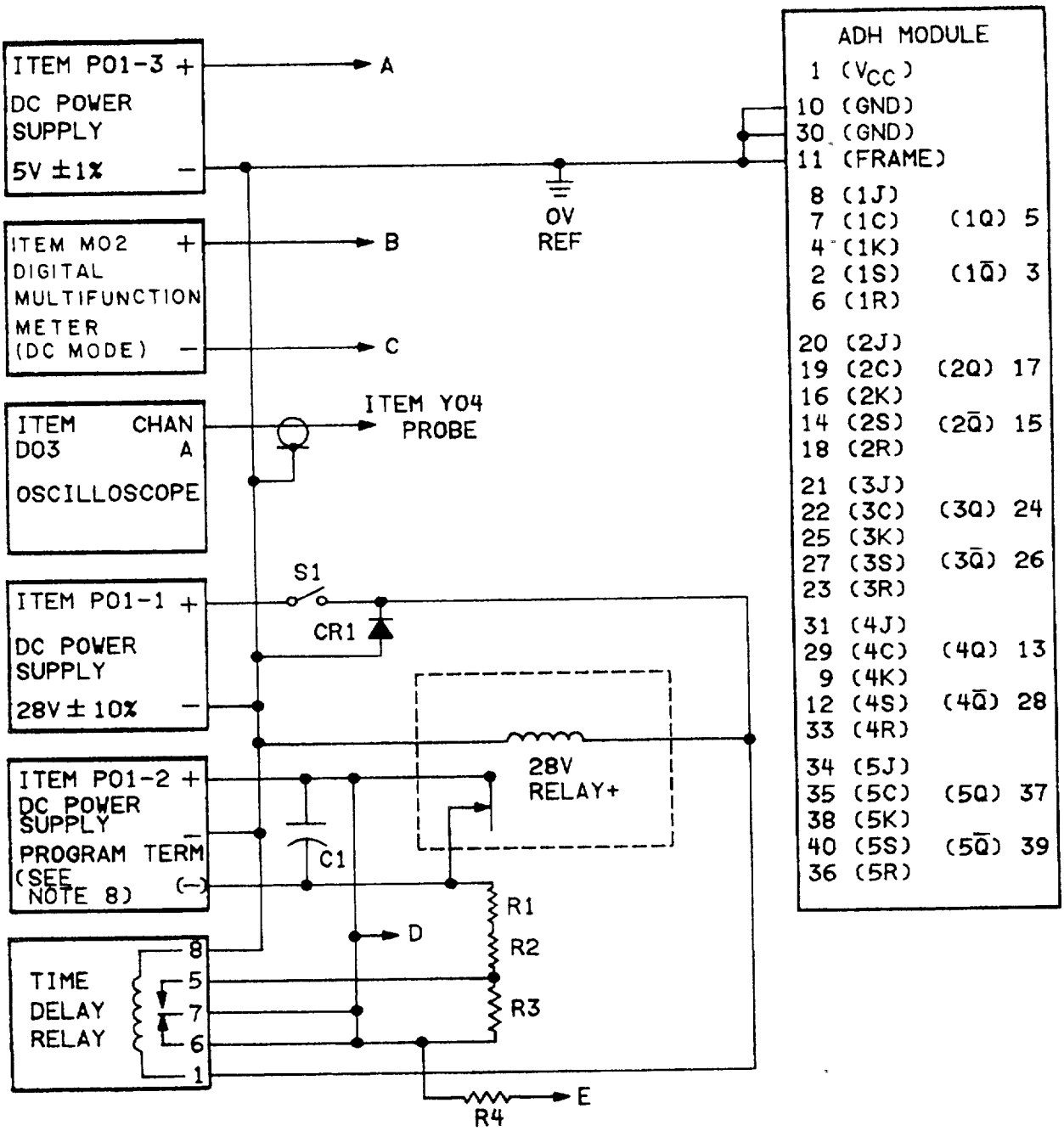
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
A14	3Q Zener applied potential, cond A	Open	24	27	1	23	3.8	5.5	V	6,7
A15	$3\overline{Q}$		26	23		27				
A16	4Q		13	12		31				
A17	4Q					29				
A18	4Q					9				
A19	4Q					33				
A20	$4\overline{Q}$		28	33		12				
A21	5Q		37	40		34				
A22	5Q					35				
A23	5Q					38				
A24	5Q					36				
A25	5Q		39	36		40				

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TABLE III. Zener applied potential (cond A and cond B) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
B1	$1\overline{Q}$ Zener applied potential, cond B	1	3	6	Open	7	3.8	5.5	V	6,7
B2	$2\overline{Q}$		15	18		19				
B3	$3\overline{Q}$		26	23		22				
B4	$4\overline{Q}$		28	33		29				
B5	$5\overline{Q}$		39	36		35				

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CAUTION

If test conditions A and B are successively repeated, the time interval between execution of each test condition shall be a minimum of 10 seconds in order to preclude detrimental effects due to module heating.

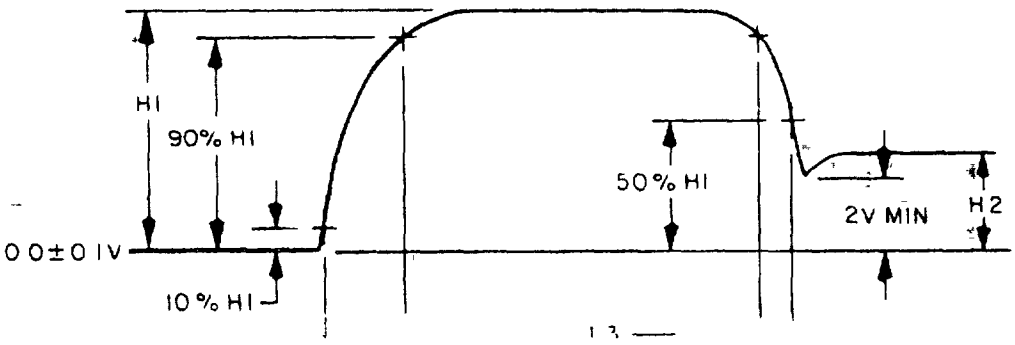
FIGURE 6. Zener applied potential (condition A and condition B) test circuit for table III.

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NOTES
1. R1 = 178Ω ±1%, 1/8 W.
2. R2 = 301Ω ±1%, 1/8 W.
3. R3 = 649Ω ±1%, 1/8 W.
4. R4 = 1.54Ω ±1%, 1/8 W.
5. C1 = 22 µF ±10%, 35 Vdc.
6. CR1 = 1N647 or equivalent
7. Resistors R1, R2, R3, and capacitor C1 are programming components for item PO1-2 and are based on a programming constant of 100 ohms per volt.
8. For each test condition, switch S1 is closed and voltage measurements shall be made after a minimum time of 2 seconds following the closure of switch S1 At the conclusion of each test condition, switch S1 shall be returned to the open position.
9 The circuit of this figure is drawn showing item PO1-2 with a negative reference terminal (non-commoned programming terminal labeled (-)). The circuit in this figure as shown, will not operate properly if item PO1-2 has a positive non-commoned programming reference terminal. NOTE Load current must not pass through sensing network.

FIGURE 6. Zener applied potential (condition A and condition B) test circuit for table III - Continued.

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NOTES
1. Using item D03 with item Y04 probe, adjust the time delay to obtain the above output waveform at pin 7 of the time delay relay after switch S1 is closed
2. Input pulse parameters
   H1 = 11.25 V ±1%
   H2 = 5.0 V ±10%.
   T1 = 100 $^{+100}_{-50}$ ms.
   T2 = 100 $^{+400}_{-50}$ ms
   T3 = 0.5 s±10%

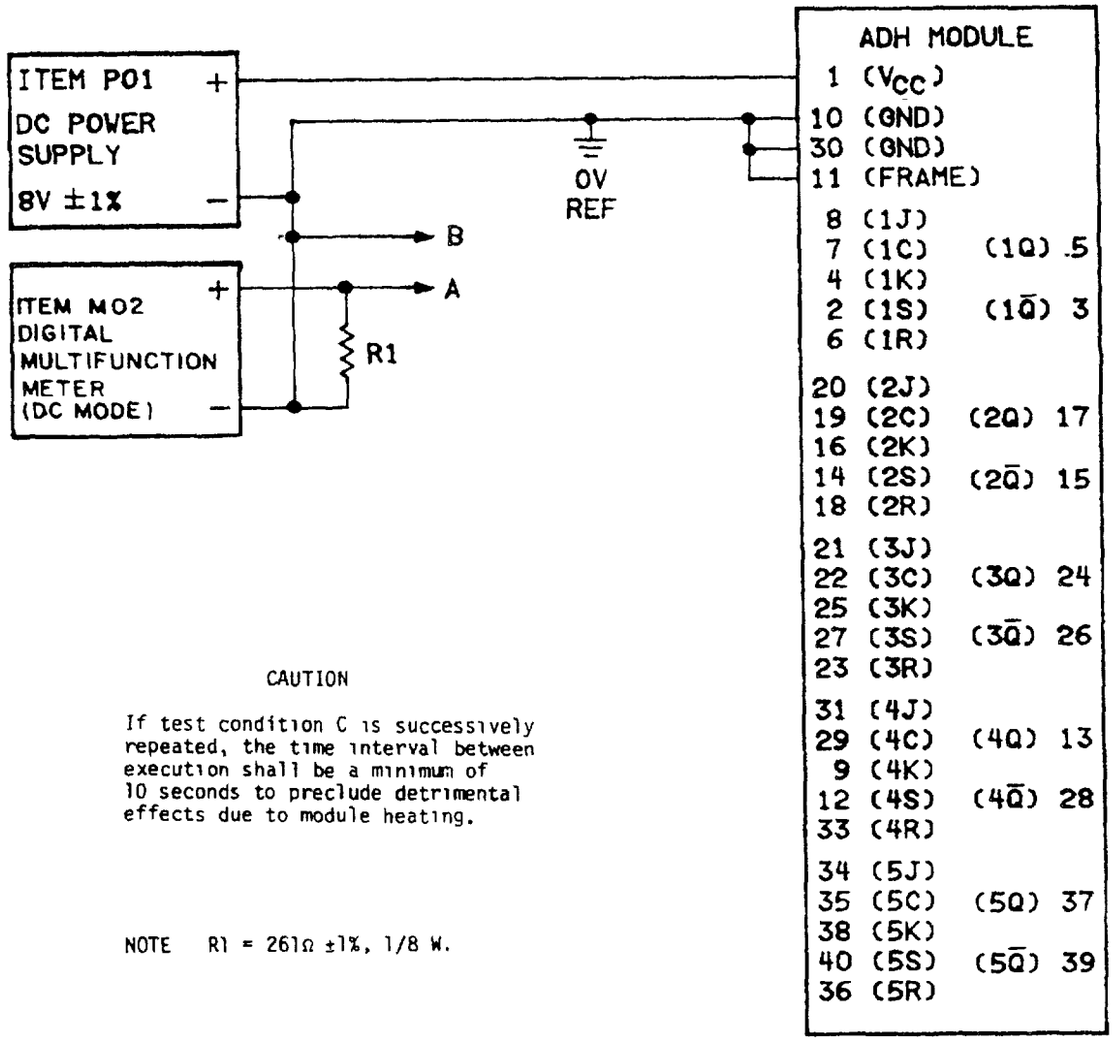
FIGURE 7. Zener applied potential (condition A and condition B) test circuit waveform for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1Q V_{OUT} (low loaded)	5	6	0.0	1.1	V	8
C2	2Q	17	18				
C3	3Q	24	23				
C4	4Q	13	33				
C5	5Q	37	36				
C6	$1\overline{Q}$	3	2				
C7	$2\overline{Q}$	15	14				
C8	$3\overline{Q}$	26	27				
C9	$4\overline{Q}$	28	12				
C10	$5\overline{Q5}$	39	40				

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CAUTION

If test condition C is successively repeated, the time interval between execution shall be a minimum of 10 seconds to preclude detrimental effects due to module heating.

NOTE R1 = 261Ω ±1%, 1/8 W.

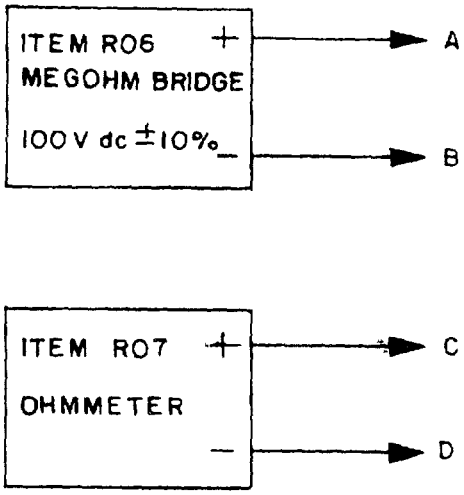
FIGURE 8. V_{OUT} (low-loaded) test circuit for table IV

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TABLE V Isolation, resistance, and continuity test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D	Isolation	11	All other pins			10	-	MΩ	9
E	Resistance	↓	↓	1	32	4	5	kΩ	↓
F1	Continuity			10	30	0	1	Ω	
F2	Continuity			11	Frame	0	1	Ω	

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE. For test condition F2, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

ADH MODULE			
1	(V_{CC})		
10	(GND)		
30	(GND)		
11	(FRAME)		
8	(1J)		
7	(1C)	(1Q)	5
4	(1K)		
2	(1S)	($1\bar{Q}$)	3
6	(1R)		
20	(2J)		
19	(2C)	(2Q)	17
16	(2K)		
14	(2S)	($2\bar{Q}$)	15
18	(2R)		
21	(3J)		
22	(3C)	(3Q)	24
25	(3K)		
27	(3S)	($3\bar{Q}$)	26
23	(3R)		
31	(4J)		
29	(4C)	(4Q)	13
9	(4K)		
12	(4S)	($4\bar{Q}$)	28
33	(4R)		
34	(5J)		
35	(5C)	(5Q)	37
38	(5K)		
40	(5S)	($5\bar{Q}$)	39
36	(5R)		

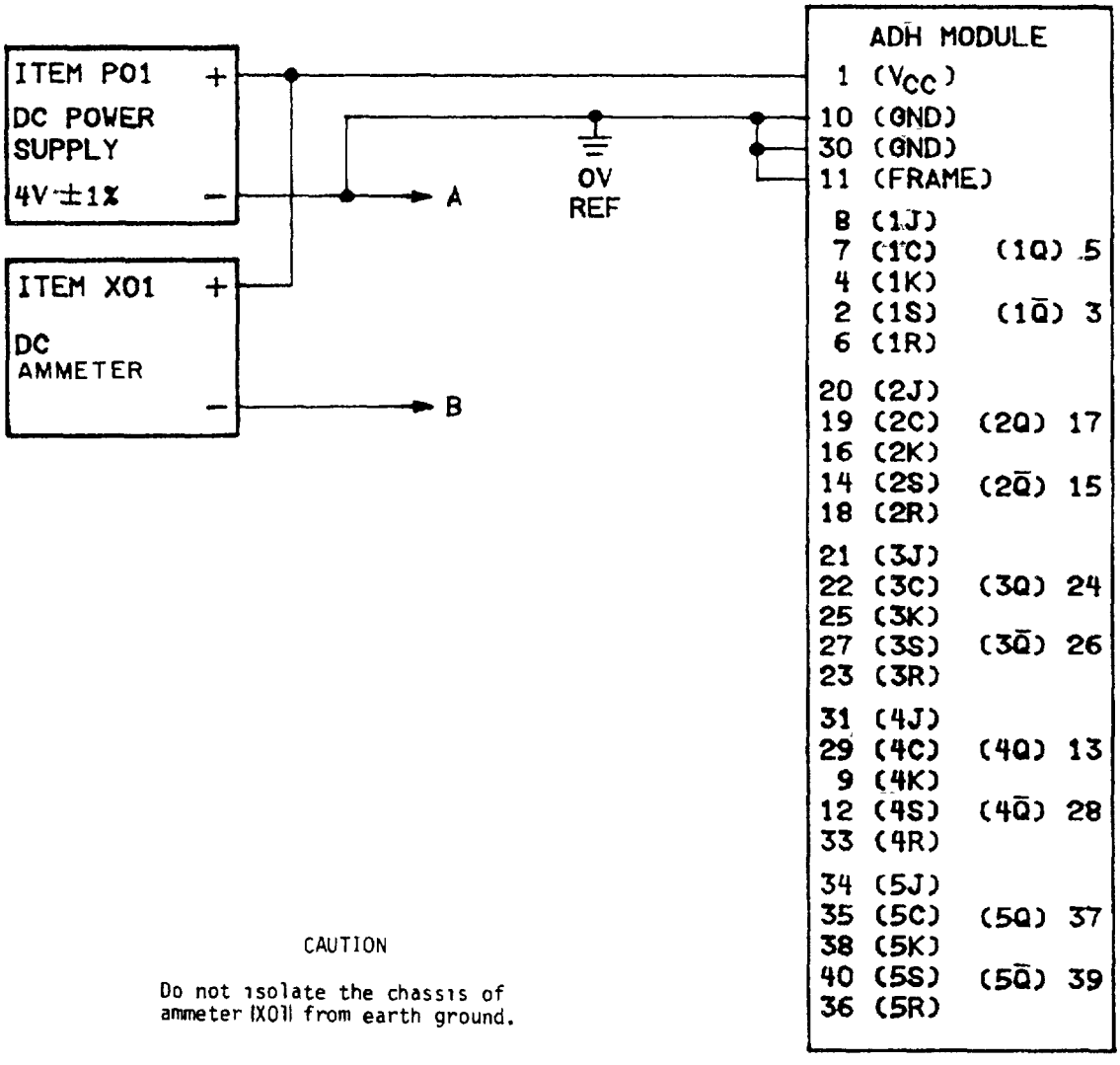
FIGURE 9. Isolation, resistance, and continuity test circuit for table V.

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TABLE VI. Input current (reverse clock) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	1C Input current, reverse clock	8, 4	7	0	4	µA	10
G2	2C	20, 16	19				
G3	3C	21, 25	22				
G4	4C	31, 9	29				
G5	5C	34, 38	35				

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CAUTION

Do not isolate the chassis of ammeter [X01] from earth ground.

FIGURE 10. Input current (reverse clock) test circuit for table VI.

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TABLE VII. Input reverse current (synchronous and asychronous) test.

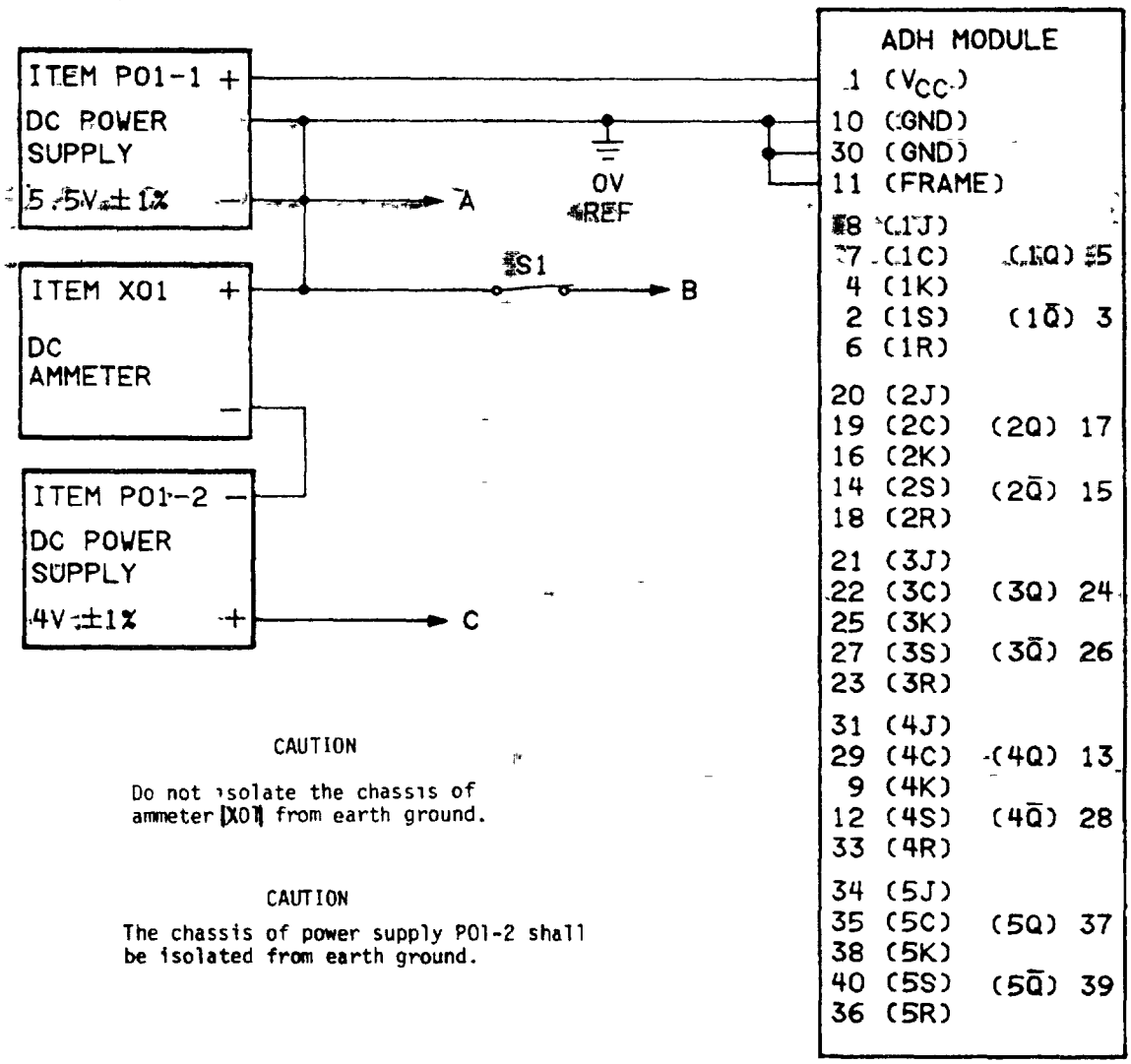
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1J Input reverse current, synchronous	7	Open	8	0	2	µA	11
H2	2J	19		20				
H3	3J	22		21				
H4	4J	29		31				
H5	5J	35		34				
H6	1K	7		4				
H7	2K	19		16				
H8	3K	22		25				
H9	4K	29		9				
H10	5K	35		38				
J1	1R Input reverse current, asynchronous	8	7	6		4		

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TABLE VII. Input reverse current (synchronous and asychronous) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
J2	2R Input reverse current, asynchronous	20	19	18	0	4	µA	11
J3	3R	21	22	23				
J4	4R4	31	29	33				
J5	5R	34	35	36				
J6	1S	4	7	2				
J7	2S	16	19	14				
J8	3S	25	22	27				
J9	4S	9	29	12				
J10	5S5	38	35	40				

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CAUTION

Do not isolate the chassis of ammeter (X01) from earth ground.

CAUTION

The chassis of power supply P01-2 shall be isolated from earth ground.

NOTE: After making all circuit connections for a test condition, open switch (S1) for at least 100 ns, then close it and make measurement.

FIGURE 11. Input reverse current (synchronous and asynchronous) test circuit for table VII.

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TABLE VIII. Input forward current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1J Input forward current	2, 6, 4	8	Open	0 20	1.07	mA	12
K2	2J	14, 18, 16	20					
K3	3J	27, 23, 25	21					
K4	4J	12, 33, 9	31					
K5	5J	40, 36, 38	34					
K6	1K	2, 6, 8	4					
K7	2K	14, 18, 20	16					
K8	3K	27, 23, 21	25					
K9	4K	12, 33, 31	9					
K10	5K	40, 36, 34	38					

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TABLE VIII. Input forward current test - Continued.

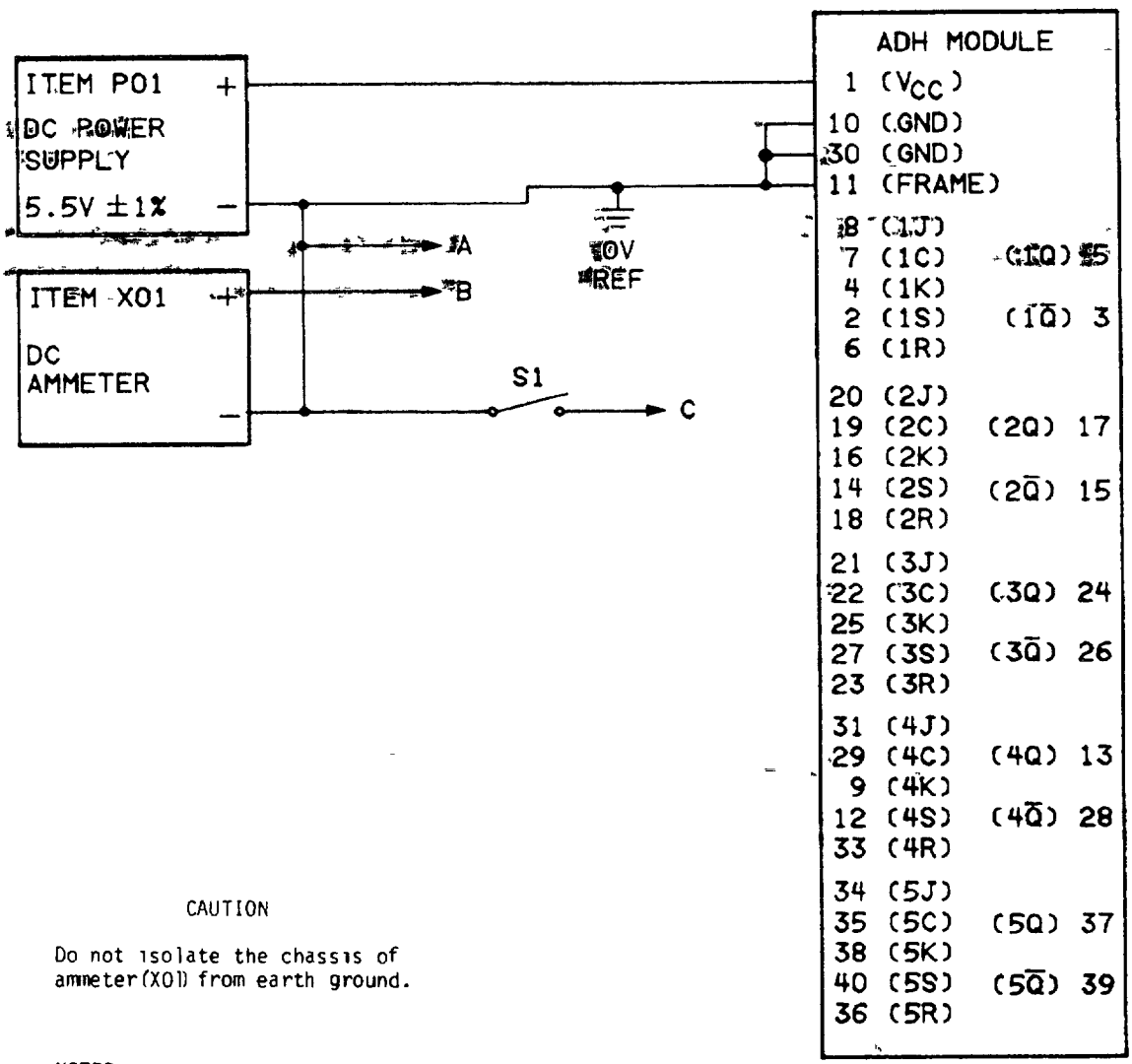
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L1	1R Input forward current	4,2	6	Open	0.2	3.2	mA	12
L2	2R	14,16	18					
L3	3R	27,35	23					
L4	4R	12,9	33					
L5	5R	40,38	36					
L6	1S	6,8	2					
L7	2S	20,18	14					
L8	3S	21,18	27					
L9	4S	31,33	12					
L10	5S	34,36	40					
M1	1C-1R	Open	7	6				
M2	2C-2R	Open	19	18				
M3	3C-3R	Open	22	23				

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TABLE VIII. Input forward current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M4	C4-R4 Input forward current	Open	29	33	0.2	3.2	mA	12
M5	C5-R5		35	36				
M6	C1-S1		7	2				
M7	C2-S2		19	14				
M8	C3-S3		22	27				
M9	C4-S4		29	12				
M10	C5-S5		35	40				

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CAUTION

Do not isolate the chassis of ammeter(X01) from earth ground.

NOTES:
1. S1 = SPST switch.
2. For test conditions M1-M10, after making all circuit connections for a test condition close switch S1 for at least 150 ns, then open it and make measurement.

FIGURE 12. Input forward current test circuit for table VIII.

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TABLE IX. Output voltage (high-level and low-level) test.

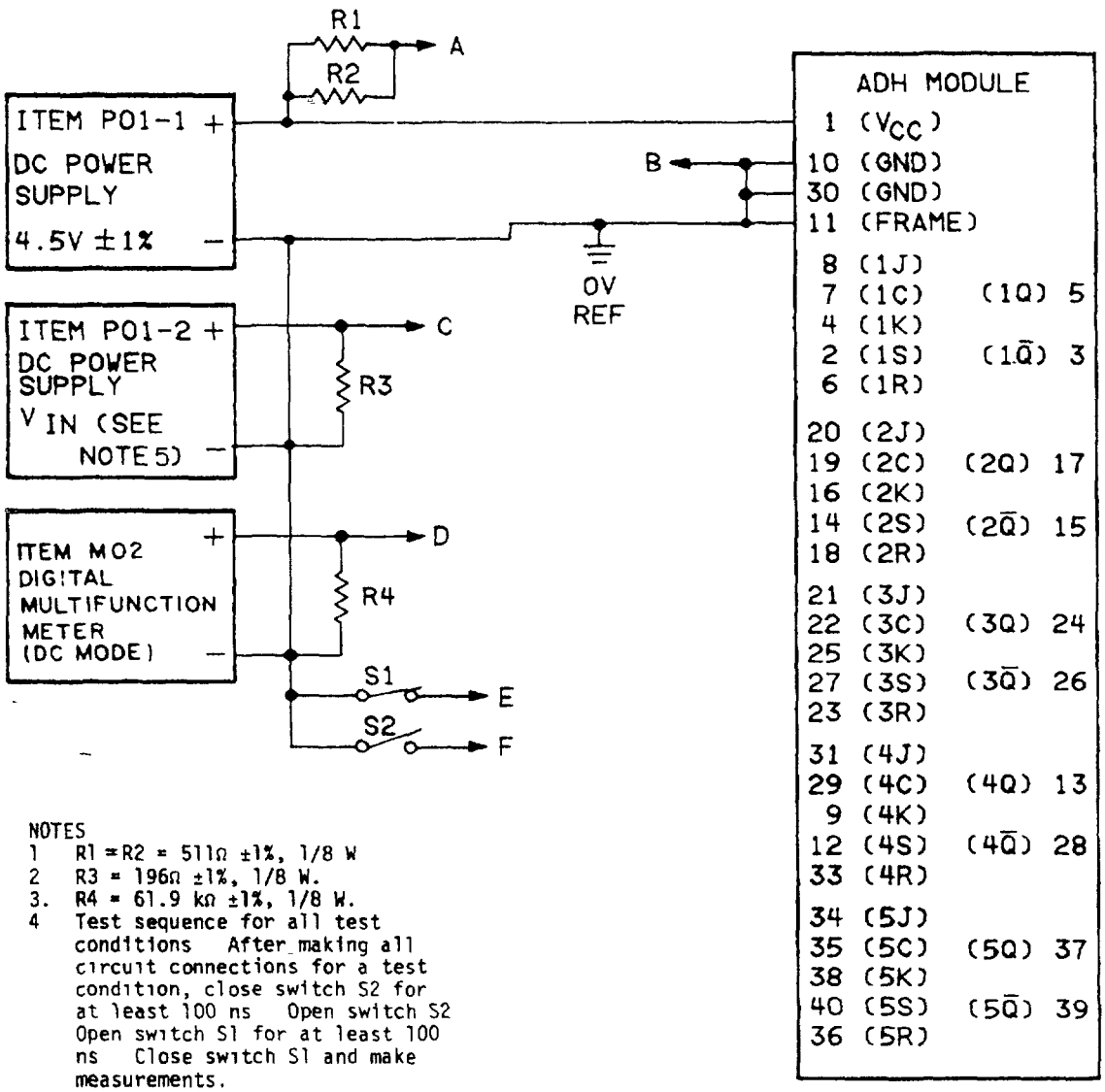
Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
N1	1Q Output voltage, high level (J input high level threshold)	Open	4	8	5	7	6	3.8	5.5	V	13
N2	2Q		16	20	17	19	18				
N3	3Q		25	21	24	22	23				
N4	4Q		9	31	13	29	33				
N5	5Q		38	34	37	35	36				
P1	$1\overline{Q}$ Output voltage, high level (K input high level threshold)		8	4	3	7	2				
P2	$2\overline{Q}$		20	16	15	19	14				
P3	$3\overline{Q}$		21	25	26	22	27				
P4	$4\overline{Q}$		31	9	28	29	12				
P5	$5\overline{Q}$		34	38	39	35	40				
Q1	1Q Output voltage, high level (S input low level threshold)		8	2	5	7	Open				
Q2	2Q		20	14	17	19					
Q3	3Q		21	27	24	22					
Q4	4Q		31	12	13	29					
Q5	5Q		34	40	37	35					

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TABLE IX. Output voltage (high-level and low-level) test - Continued

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
R1	$1\overline{Q}$ Output voltage, high level (R input low level threshold)	Open	4	6	3	7	Open	3.8	5.5	V	13
R2	$2\overline{Q}$		16	18	15	19					
R3	$3\overline{Q}$		25	23	26	22					
R4	$4\overline{Q}$		9	33	28	29					
R5	$5\overline{Q}$		38	36	39	35					
S1	1Q Output voltage, low level (S input high level threshold)	5	Open	2	5	Open	6	0.00	0.40		
S2	2Q	17		14	17		18				
S3	3Q	24		27	24		23				
S4	4Q	13		12	13		33				
S5	5Q	37		40	37		36				
T1	$1\overline{Q}$ Output voltage, low level (R input high level threshold)	3		6	3		2				
T2	$2\overline{Q}$	15		18	15		14				
T3	$3\overline{Q}$	26		23	26		27				
T4	$4\overline{Q}$	28		33	28		12				
T5	$5\overline{Q}$	39		36	39		40				

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NOTES
1. R1 = R2 = 511Ω ±1%, 1/8 W
2. R3 = 196Ω ±1%, 1/8 W.
3. R4 = 61.9 kΩ ±1%, 1/8 W.
4. Test sequence for all test conditions. After making all circuit connections for a test condition, close switch S2 for at least 100 ns. Open switch S2. Open switch S1 for at least 100 ns. Close switch S1 and make measurements.
5. Use appropriate V_{IN} for specified test condition and temperature listed below

Test condition	Temperature (°C)	V_{IN} (volts)
N1-N5, P1-P5, S1-S5, T1-T5	0	2.0
	25	1 9
	60	1 8
	80	1.8
Q1-Q5, R1-R5	0	1.20
	25	1.10
	60	0 95
	80	0 80

0°, 60°, and 80°C refer to operating temperatures (module frame), 25°C refers to ambient temperature. For 80°C use the 0° to 60°C end-of-life limits."

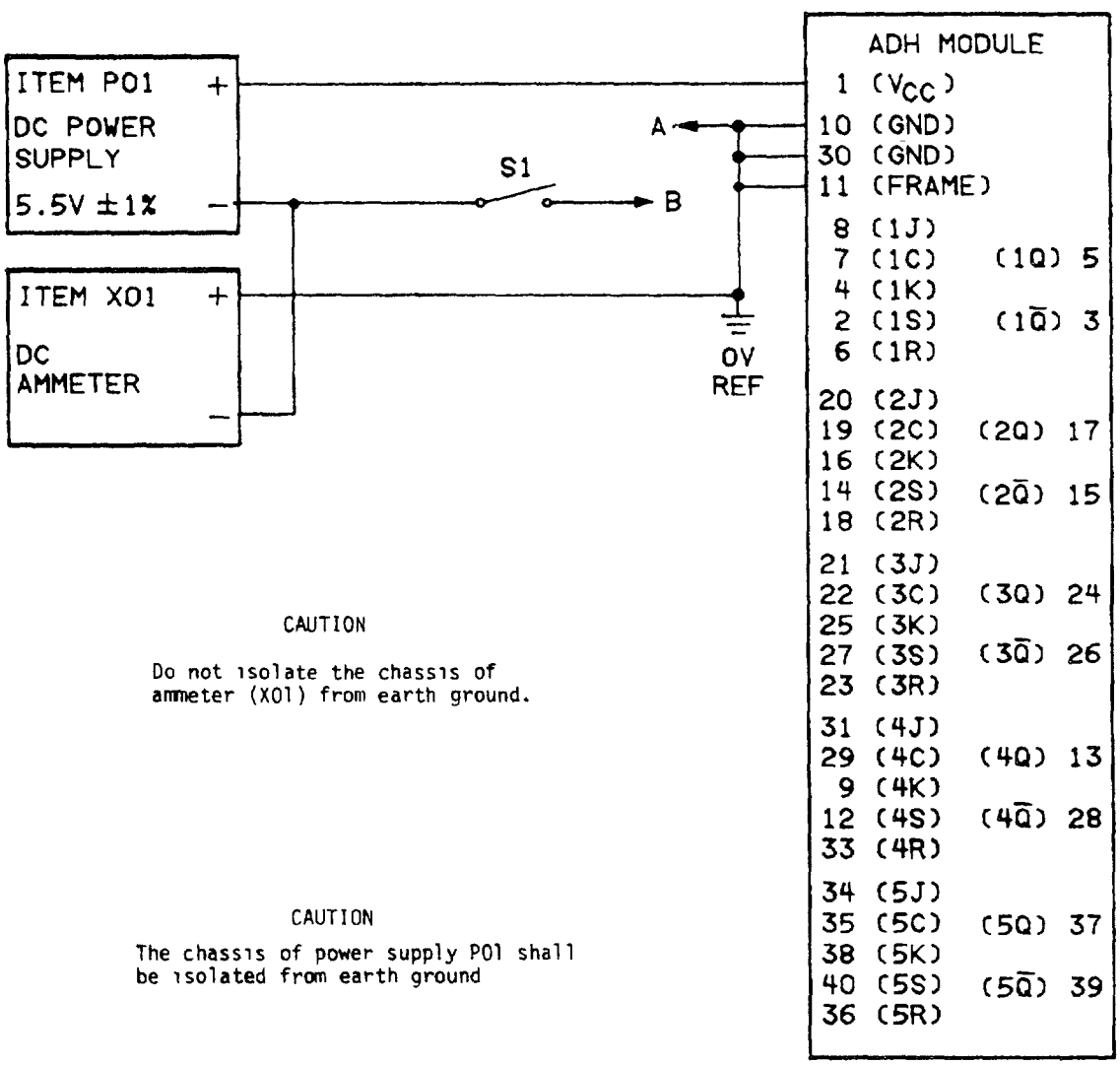
FIGURE 13. Output voltage (high-level and low-level) test circuit for table IX.

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TABLE X. Output short circuit current and power supply current test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
U1	1Q	Output short circuit current	2	5	0.9	3.0	mA	14
U2	2Q		14	17				
U3	3Q		27	24				
U4	4Q		12	13				
U5	5Q		40	37				
U6	$1\overline{Q}$		6	3				
U7	$2\overline{Q}$		18	15				
U8	$3\overline{Q}$		23	26				
U9	$4\overline{Q}$		33	28				
U10	$5\overline{Q}$		36	39				
V1	S	Power supply current	7,19,22,29,35	2,14,27,12,40	0.0	82.5		
V2	R	Power supply current	7,19,22,29,35	6,18,23,33,36	0.0	82.5		

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CAUTION

Do not isolate the chassis of ammeter (X01) from earth ground.

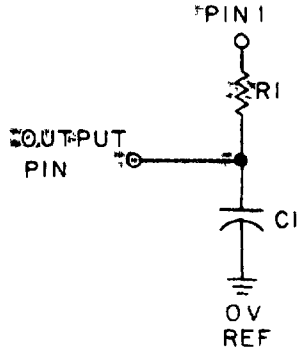
CAUTION

The chassis of power supply P01 shall be isolated from earth ground

NOTES
1. S1 = SPST switch.
2 After making all circuit connections for a test condition close switch S1 for at least 100 ns, then open it and make measurement

FIGURE 14. Output short circuit current and power supply current test circuit for table I

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NOTES
1. Load circuit parameters
   (a) Load circuit 1
       R1 = 3.83 kΩ ±1%, 1/8 W.
       C1 = 25 pF minimum, 200 pF maximum (see note 2)
   (b) Load circuit 2
       R1 = 3.83 kΩ ±1%, 1/8 W.
       C1 = 30 ±1 pF (see note 2)
   (c) Load circuit 3
       R1 = 332Ω ±5 pF (see note 2)
       C1 = 30 ±5 pF (see note 2)
   (d) Load circuit 4
       R1 = 3.83 kΩ ±1%, 1/8 W
       C1 = 30 ±5 pF (see note 2)

2. With the module removed from the test connector and using item C02, measure capacitance (C1) in notes 1(a), 1(b), 1(c), and 1(d). The measurements shall be performed at the point of mechanical contact between the module blade pin and the test connector pin. The measured values (stray and compensating capacitance) at 1 MHz ±10%, plus the probe capacitance shall be as specified. The resistors shall be disconnected (as close to the body of the resistor as practical) at the test connector pins and then reconnected prior to using the test connector for module testing.
3. The loop length from pin 1 to each load resistor and output pin, including the length of the body of the resistor, shall not exceed 4.5 inches (114.3 mm).

FIGURE 15 ADH logic load circuit

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TABLE XI. Output voltage (high-level) test.

Test condition	Parameter	Connect						Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
W1	$1\overline{Q}$ Output voltage, high level (J input low level threshold)	2	6	8	7	3	4	3.8	5.5	V	16,17
W2	$2\overline{Q}$	14	18	20	19	15	16				
W3	$3\overline{Q}$	27	23	21	22	26	25				
W4	$4\overline{Q}$	12	33	31	29	28	9				
W5	$5\overline{Q}$	40	36	34	35	39	38				
X1	1Q Output voltage, high level (K input low level threshold)	6	2	4	7	5	8				
X2	2Q	18	14	16	19	17	20				
X3	3Q	23	27	25	22	24	21				
X4	4Q	33	12	9	29	13	31				
X5	5Q	36	40	38	35	37	34				

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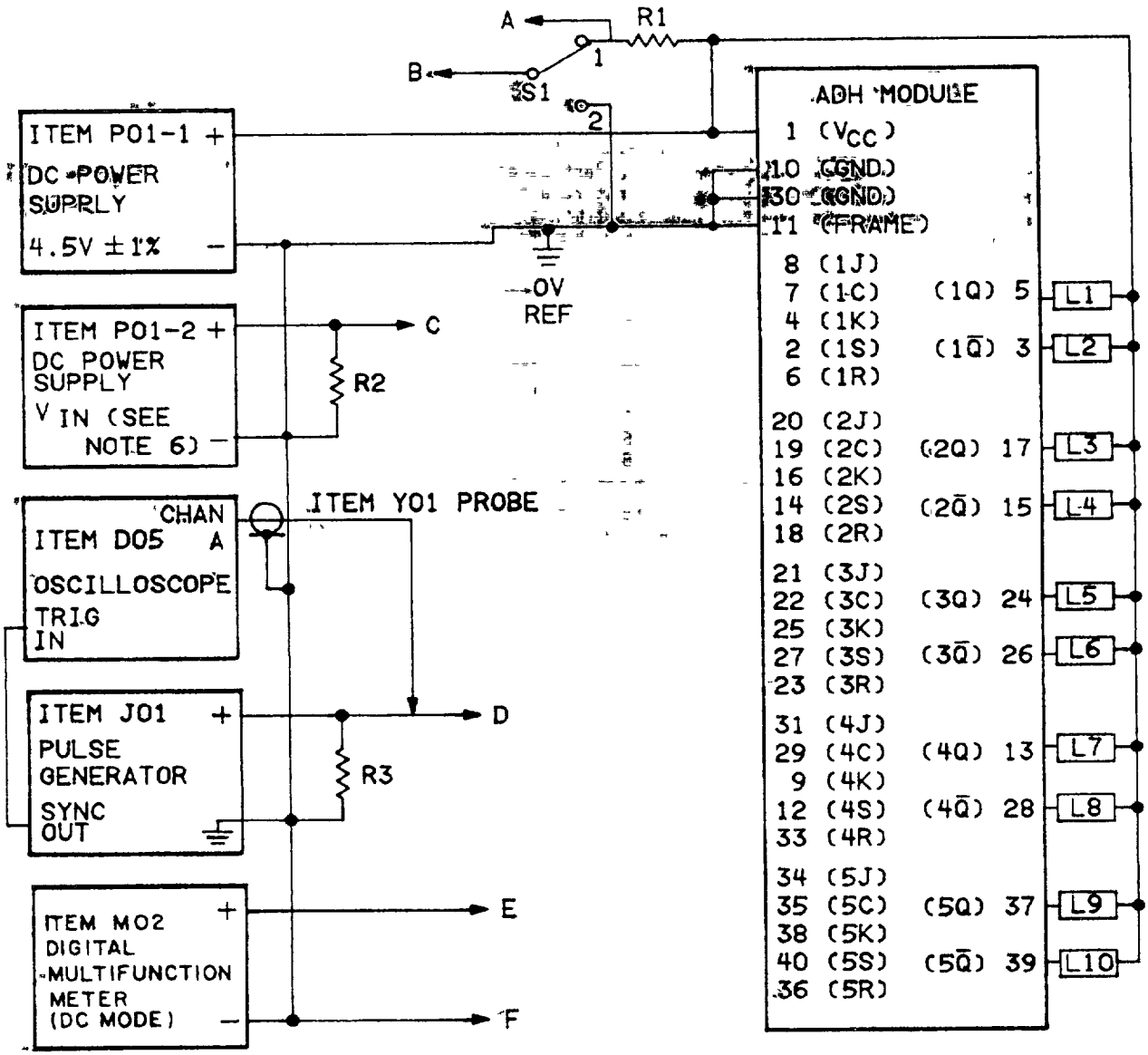


FIGURE 16. Output voltage (high-level) test circuit for table XI.

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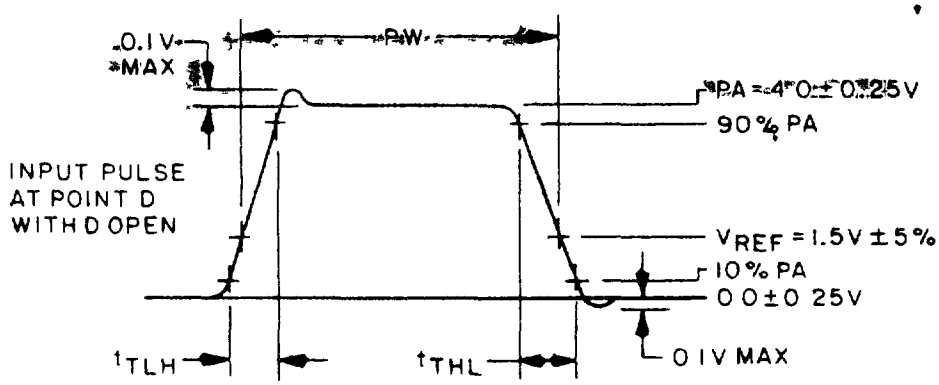
NOTES
1. R1 = 5.9 kΩ ±1%, 1/8 W.
2. R2 = 196Ω ±1%, 1/8 W.
3. R3 = 50Ω ±5%, 1 W.
4. L1-L10 = Use load circuit 1 as shown on figure 15.
5. After making all circuit connections for a test condition, rotate switch S1 to position 2 for at least 100 ns, rotate switch S1 back to position 1 and make output voltage measurements.
6. Use appropriate V_{IN} for specified test condition and temperature listed below

Test condition	Temperature (°C)	V_{IN} (volts)
W1-W5, X1-X5	0	1.20
	25	1.10
	60	0.95
	80	0.80

0°, 60°, and 80°C refer to operation temperatures (module frame); 25°C refers to ambient temperature. For 80°C use the 0° to 60°C end-of-life limits."

FIGURE 16. Output voltage (high-level) test circuit for table XI - Continued.

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NOTE Input pulse parameter
PW = 2.0 µs ±10%.
$t_{TLH} = t_{THL}$ = 500 ns ±10%.
PRR = 100 kHz ±10%

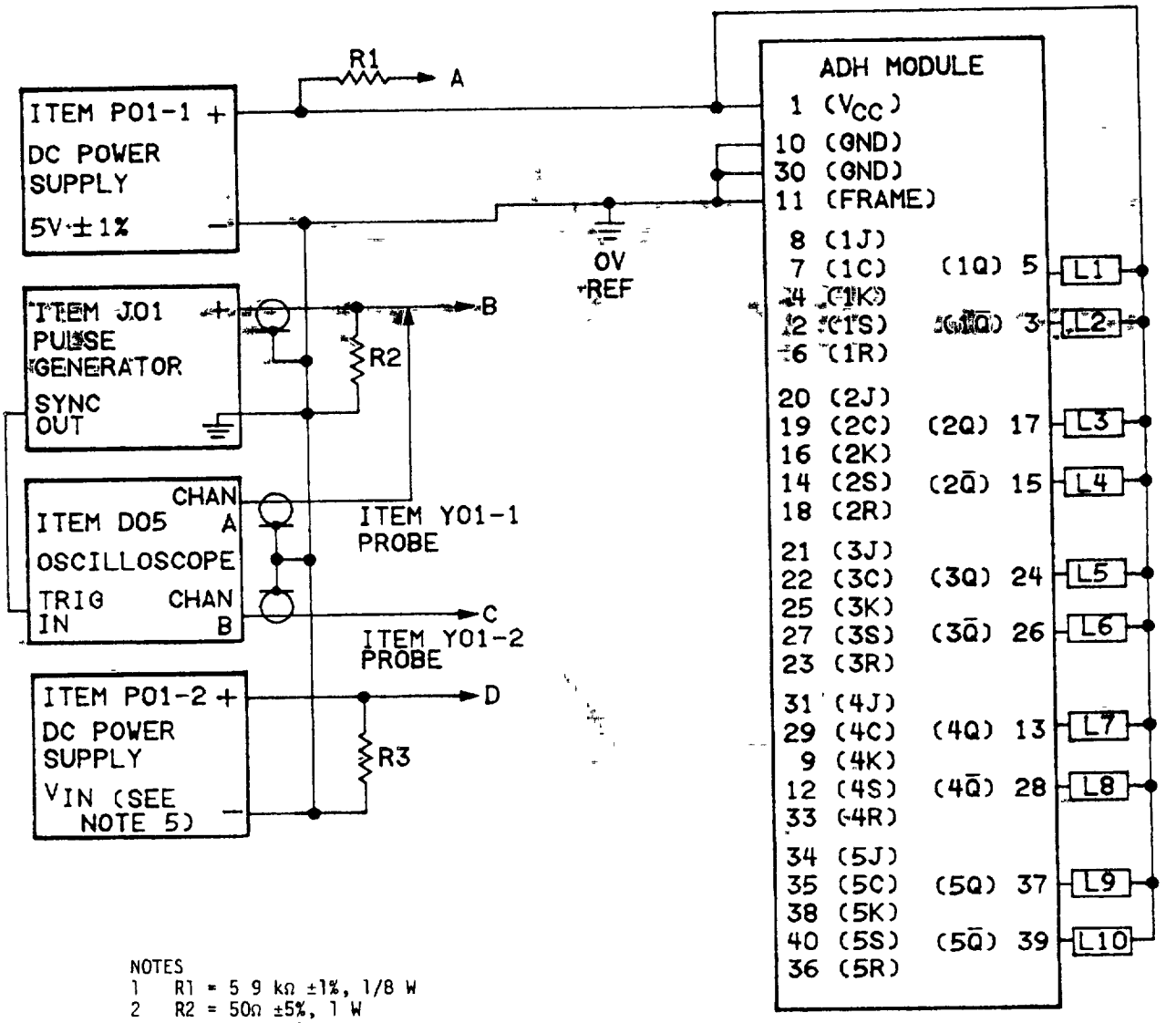
FIGURE 17. Output voltage (high-level) test circuit waveform for table XI.

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TABLE XII. Operation (J-K) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y1	J1-K1 Operation, J-K	2,6	7	5	8,4	16	24	µs	18,19
Y2	J2-K2	14,18	19	17	20,16	↓	↓	↓	↓
Y3	J3-K3	27,23	22	24	21,25				
Y4	J4-K4	12,33	29	13	31,9				
Y5	J5-K5	40,36	35	37	34,38				

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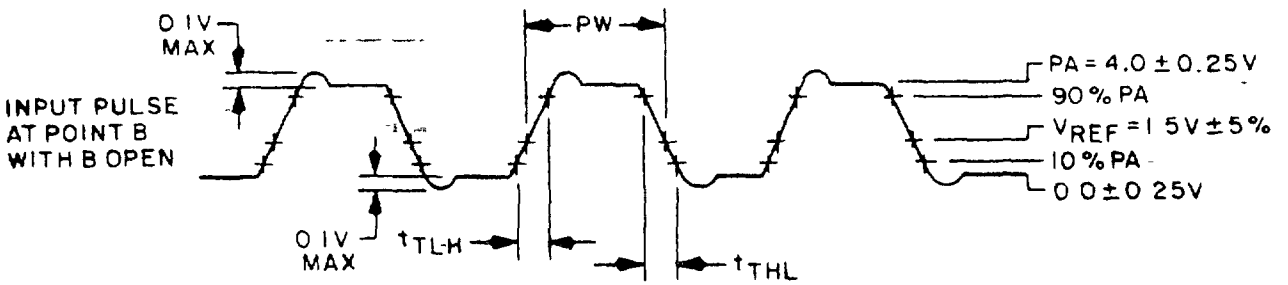
NOTES
1 R1 = 5 9 kΩ ±1%, 1/8 W
2 R2 = 50Ω ±5%, 1 W
3 R3 = 196Ω ±1%, 1/8 W.
4 L1-L10 = Use load circuit 1 as shown on figure 15.
5 Use appropriate V_{IN} for specified test condition and temperature listed below

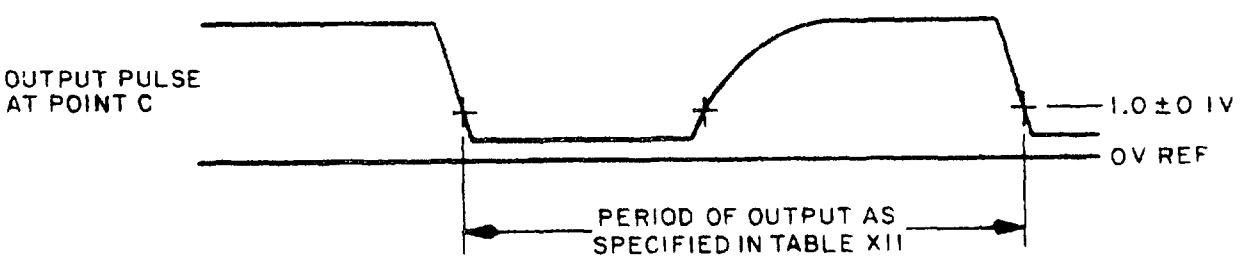
Test condition	Temperature (°C)	V_{IN} (volts)
Y1-Y5	0	2.0
	25	1.9
	60	1.8
	80	1.8

0°, 60°, and 80°C refer to operating temperatures (module frame), 25°C refers to ambient temperature For 80°C use the 0° to 60°C end-of-life limits.

FIGURE 18. Operation (J-K) test circuit for table XII.

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NOTE Input pulse parameters
PA = 4.0 ±0.25V
$t_{TLH} = t_{THL}$ = 500 ±50 ns.
PW = 2 µs ±10%
PRR = 100 kHz ±10%.

FIGURE 19. Operation (J-K) test circuit waveforms for table XII.

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TABLE XIII. Parallel output (#1 - #4) test

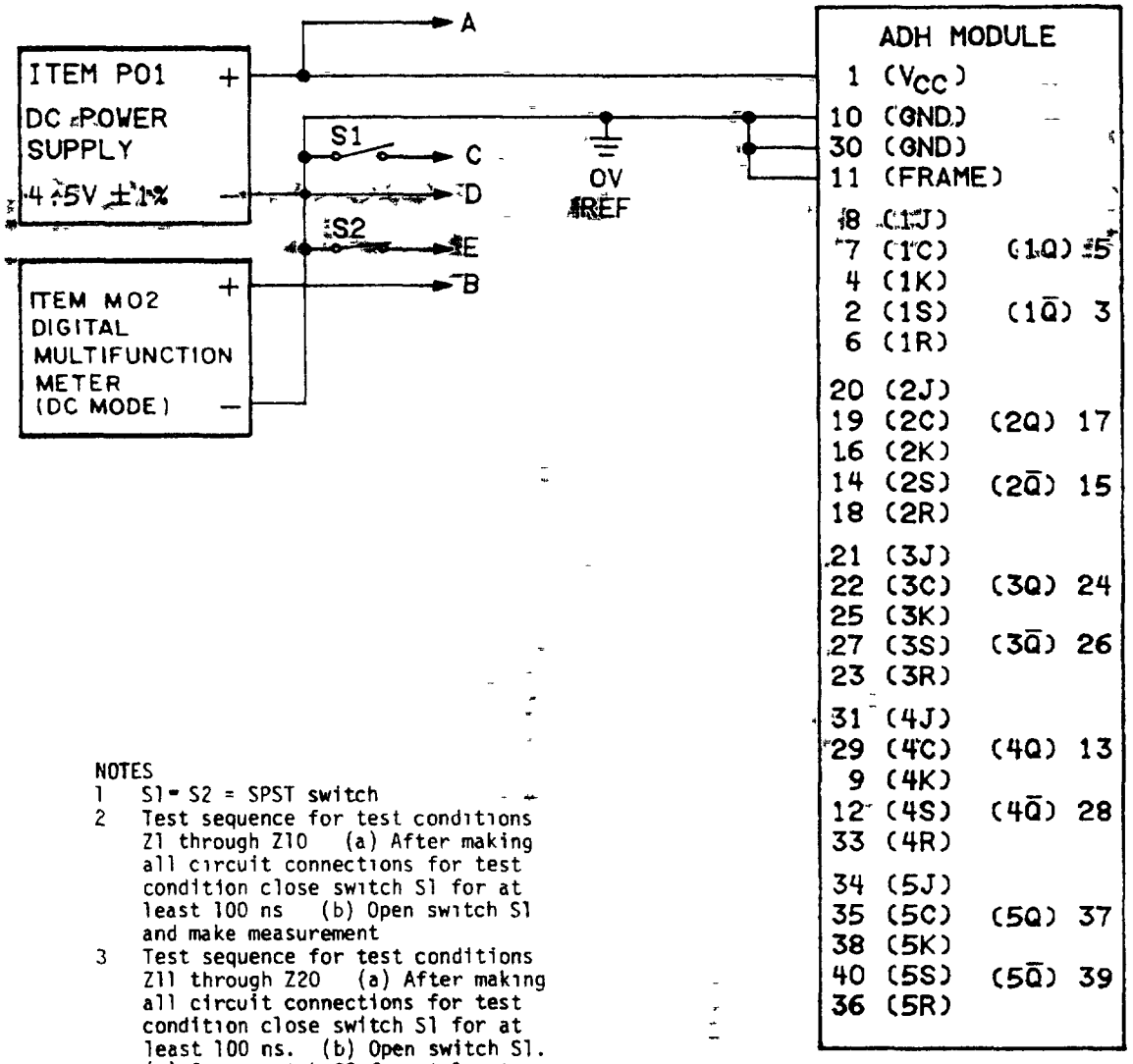
Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z1	1Q Parallel output #1	Open	5	6	3	Open	0 00	0 40	V	20
Z2	2Q		17	18	15					
Z3	3Q		24	23	26					
Z4	4Q		13	33	28					
Z5	5Q		37	36	39					
Z6	1$\overline{Q}$ Parallel output #2		3	2	5					
Z7	2$\overline{Q}$		15	14	17					
Z8	3$\overline{Q}$		26	27	24					
Z9	4$\overline{Q}$		28	12	13					
Z10	5$\overline{Q}$		39	40	37					
Z11	1Q Parallel output #3	8	5	6	3, 4	7				
Z12	2Q Parallel output #3	20	17	18	15, 16	19				

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TABLE XIII. Parallel output (#1 - #4) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z13	3Q Parallel output #3	21	24	23	26, 25	22	0.00	0.40	V	20
Z14	4Q Parallel output #3	31	13	33	28, 9	29				
Z15	5Q Parallel output #3	34	37	36	39, 38	35				
Z16	$1\overline{Q}$ Parallel output #4	4	3	2	5, 8	7				
Z17	$2\overline{Q}$	16	15	14	17, 20	19				
Z18	$3\overline{Q}$	25	26	27	24, 21	22				
Z19	$4\overline{Q}$	9	28	12	13, 31	29				
Z20	$5\overline{Q}$	38	39	40	37, 34	35				

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NOTES
1. S1 - S2 = SPST switch
2. Test sequence for test conditions Z1 through Z10 (a) After making all circuit connections for test condition close switch S1 for at least 100 ns (b) Open switch S1 and make measurement
3. Test sequence for test conditions Z11 through Z20 (a) After making all circuit connections for test condition close switch S1 for at least 100 ns. (b) Open switch S1. (c) Open switch S2 for at least 100 ns. (d) Close switch S2 and make measurement.

FIGURE 20. Parallel output (#1 - #4) test circuit for table XIII.

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TABLE XIV. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA1	1Q Transition time (t_{TLH})	7, 8, 4	5	55	170	ns	21,22
AA2	2Q	19,20,16	17				
AA3	3Q	22,21,25	24				
AA4	4Q	29,31, 9	13				
AA5	5Q	35,34,38	37				
AA6	$1\overline{Q}$	7, 8, 4	3				
AA7	$2\overline{Q}$	19,20,16	15				
AA8	$3\overline{Q}$	22,21,25	26				
AA9	$4\overline{Q}$	29,31, 9	28				
AA10	$5\overline{Q}$	35,34,38	39				
AA11	1Q Transition time (t_{THL})	7, 8, 4	5	6	24		
AA12	2Q	19,20,16	17				
AA13	3Q	22,21,25	24				
AA14	4Q	29,31, 9	13				
AA15	5Q	35,34,38	37				
AA16	$1\overline{Q}$	7, 8, 4	3				
AA17	$2\overline{Q}$	19,20,16	15				
AA18	$3\overline{Q}$	22,21,25	26				
AA19	$4\overline{Q}$	29,31, 9	28				
AA20	$5\overline{Q}$	35,34,38	39				

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TABLE XV. Propagation delay time (t_{PHL} synchronous and t_{PLH} synchronous) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB1	1Q Propagation delay time, t_{PHL} synchronous	7, 8, 4	5	20	80	ns	21,22
BB2	2Q	19,20,16	17				
BB3	3Q	22,21,25	24				
BB4	4Q	29,31, 9	13				
BB5	5Q	35,34,38	37				
BB6	$1\overline{Q}$	7, 8, 4	3				
BB7	$2\overline{Q}$	19,20,16	15				
BB8	$3\overline{Q}$	22,21,25	26				
BB9	$4\overline{Q}$	29,31, 9	28				
BB10	$5\overline{Q}$	35,34,38	39				
BB11	1Q Propagtion delay time, t_{PLH} synchronous	7, 8, 4	5	30	90		
BB12	2Q	19,20,16	17				
BB13	3Q	22,21,25	24				
BB14	4Q	29,31, 9	13				
BB15	5Q	35,34,38	37				
BB16	$1\overline{Q}$	7, 8, 4	3				
BB17	$2\overline{Q}$	19,20,16	15				
BB18	$3\overline{Q}$	22,21,25	26				
BB19	$4\overline{Q}$	29,31, 9	28				
BB20	$5\overline{Q}$	35,34,38	39				

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TABLE XVI. Output voltage (high level and low level) test.

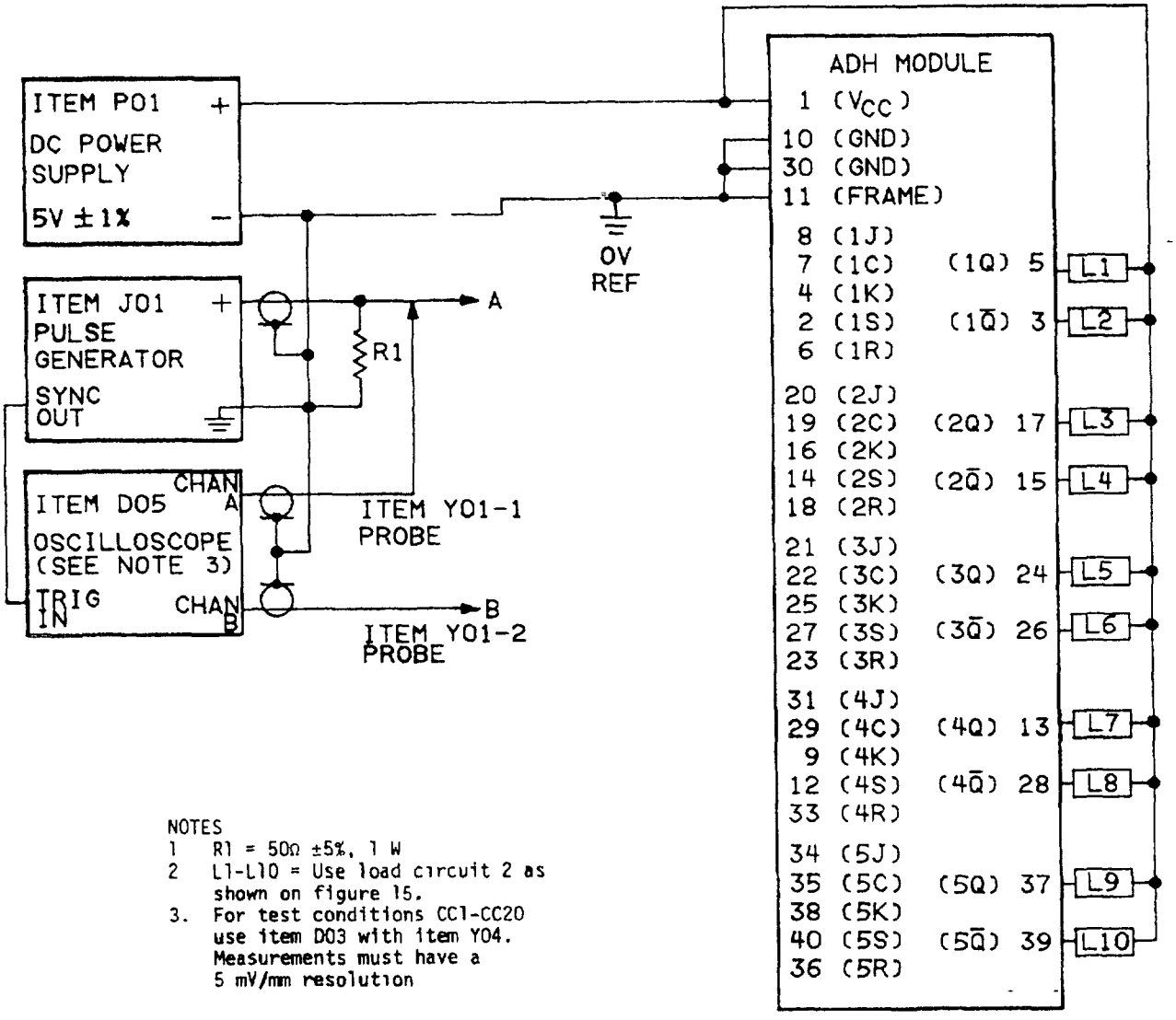
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC1	1Q Output voltage, high level	7, 8, 4	5	3.8	5.5	V	21,22
CC2	2Q	19, 20, 16	17				
CC3	3Q	22, 21, 25	24				
CC4	4Q	29, 31, 9	13				
CC5	5Q	35, 34, 38	37				
CC6	$1\overline{Q}$	7, 8, 4	3				
CC7	$2\overline{Q}$	19, 20, 16	15				
CC8	$3\overline{Q}$	22, 21, 25	26				
CC9	$4\overline{Q}$	29, 31, 9	28				
CC10	$5\overline{Q}$	35, 34, 38	39				

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TABLE XVI. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
CC11	1Q Output voltage, low level	7, 8, 4	5	0.0	0.40	V	21,22
CC12	2Q	19, 20, 16	17				
CC13	3Q	22, 21, 25	24				
CC14	4Q	29, 31, 9	13				
CC15	5Q	35, 34, 38	37				
CC16	1$\overline{Q}$	7, 8, 4	3				
CC17	2$\overline{Q}$	19, 20, 16	15				
CC18	3$\overline{Q}$	22, 21, 25	26				
CC19	4$\overline{Q}$	29, 31, 9	28				
CC20	5$\overline{Q}$	35, 34, 38	39				

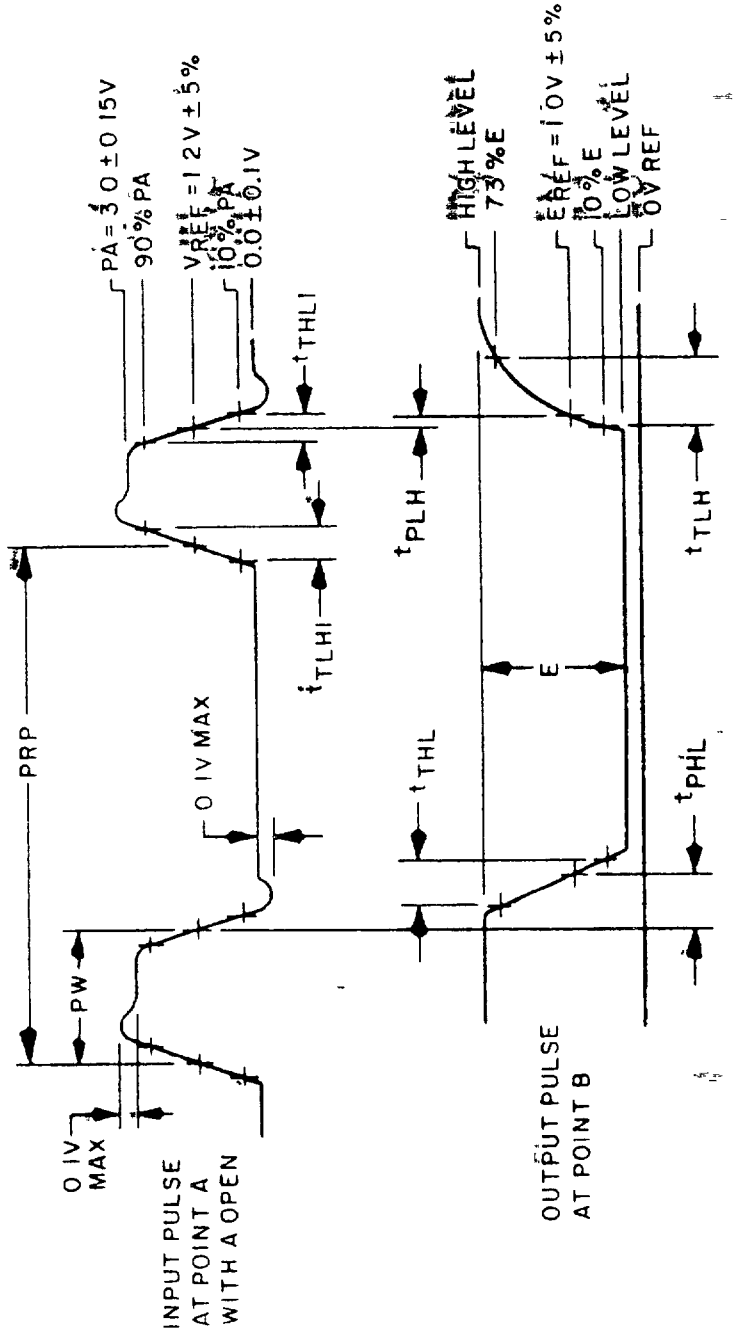
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NOTES
1. R1 = 50Ω ±5%, 1 W
2. L1-L10 = Use load circuit 2 as shown on figure 15.
3. For test conditions CC1-CC20 use item D03 with item Y04. Measurements must have a 5 mV/mm resolution

FIGURE 21. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PHL} synchronous and t_{PLH} synchronous), and output voltage (high-level and low-level) test circuit for tables XIV, XV and XVI.

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NOTE Input pulse parameters
PW = 55 ±5 ns.
PRP = 1.0 µs ±10%.
$t_{TLH1} = t_{THL1}$ = 20 ±2 ns

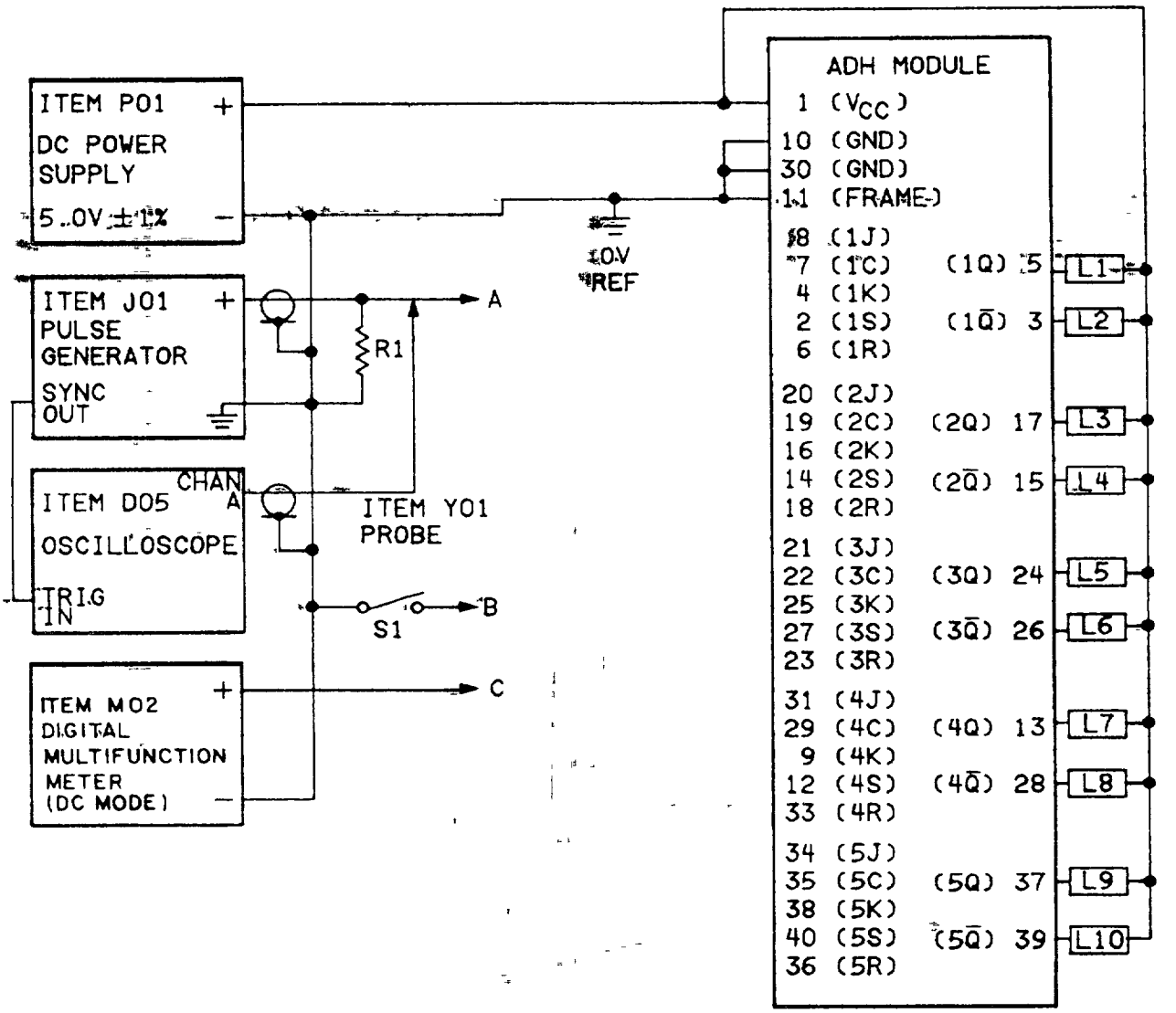
FIGURE 22 Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PHL} synchronous and t_{PLH} synchronous), and output voltage (high-level and low-level) test circuit waveforms for tables XIV, XV and XVI.

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TABLE XVII. Transient immunity (3 V and 1 5 V, clock) test

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
DD1	1C Transient immunity (3V), clock	7	2	3	0.00	0 40	V	23,24
DD2	2C	19	14	15				
DD3	3C	22	27	26				
DD4	4C	29	12	28				
DD5	5C	35	40	39				
DD6	1C Transient immunity (1.5V), clock	7	6	5				
DD7	2C	19	18	17				
DD8	3C	22	23	24				
DD9	4C	29	33	13				
DD10	5C	35	36	37				

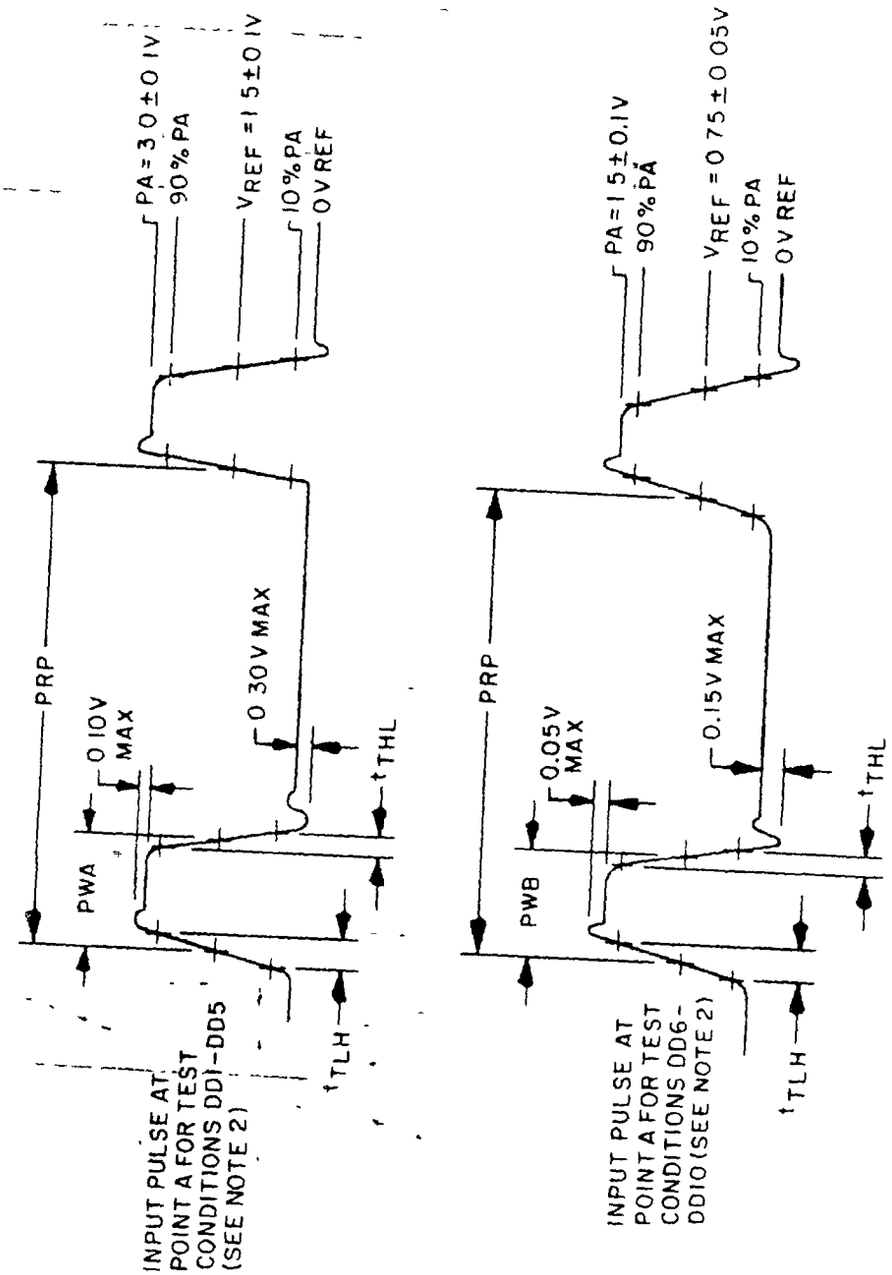
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NOTES
1. R1 = 50Ω ±5%, 1 W
2. L1-L10 = Use load circuit 4 as shown on figure 15.
3. S1 = SPST switch
4. For each test condition After making all circuit connections, close switch S1 for at least 100 ns, then open it and make measurement

FIGURE 23. Transient immunity (3 V and 1.5 V, clock) test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PWA = 10 ±0.5 ns.
   PWB = 15 ±1 ns.
   PRP = 1.0 µs ±10%.
   $t_{TLH} = t_{THL}$ = 6 ±1 ns
2. Adjust item J01 to obtain the input waveform at point A with A open. After point A has been connected some readjustment may be necessary due to loading by item J01

FIGURE 24. Transient immunity (3 V and 1.5 V, clock) test circuit waveforms for table XVII.

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TABLE XVIII. Clock threshold (high-level slow transition times and low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
EE1	1C Clock threshold, high level slow transition times	7	5	Open	4	6	ms	25,26
EE2	2C	19	17					
EE3	3C	22	24					
EE4	4C	29	13					
EE5	5C	35	37					
FF1	1C Clock threshold, low level	7	5	6	0.00	0.40	V	
FF2	2C	19	17	18				
FF3	3C	22	24	23				
FF4	4C	29	13	33				
FF5	5C	35	37	36				
FF6	1C	7	3	2				
FF7	2C	19	15	14				
FF8	3C	22	26	27				
FF9	4C	29	28	12				
FF10	5C	35	39	40				

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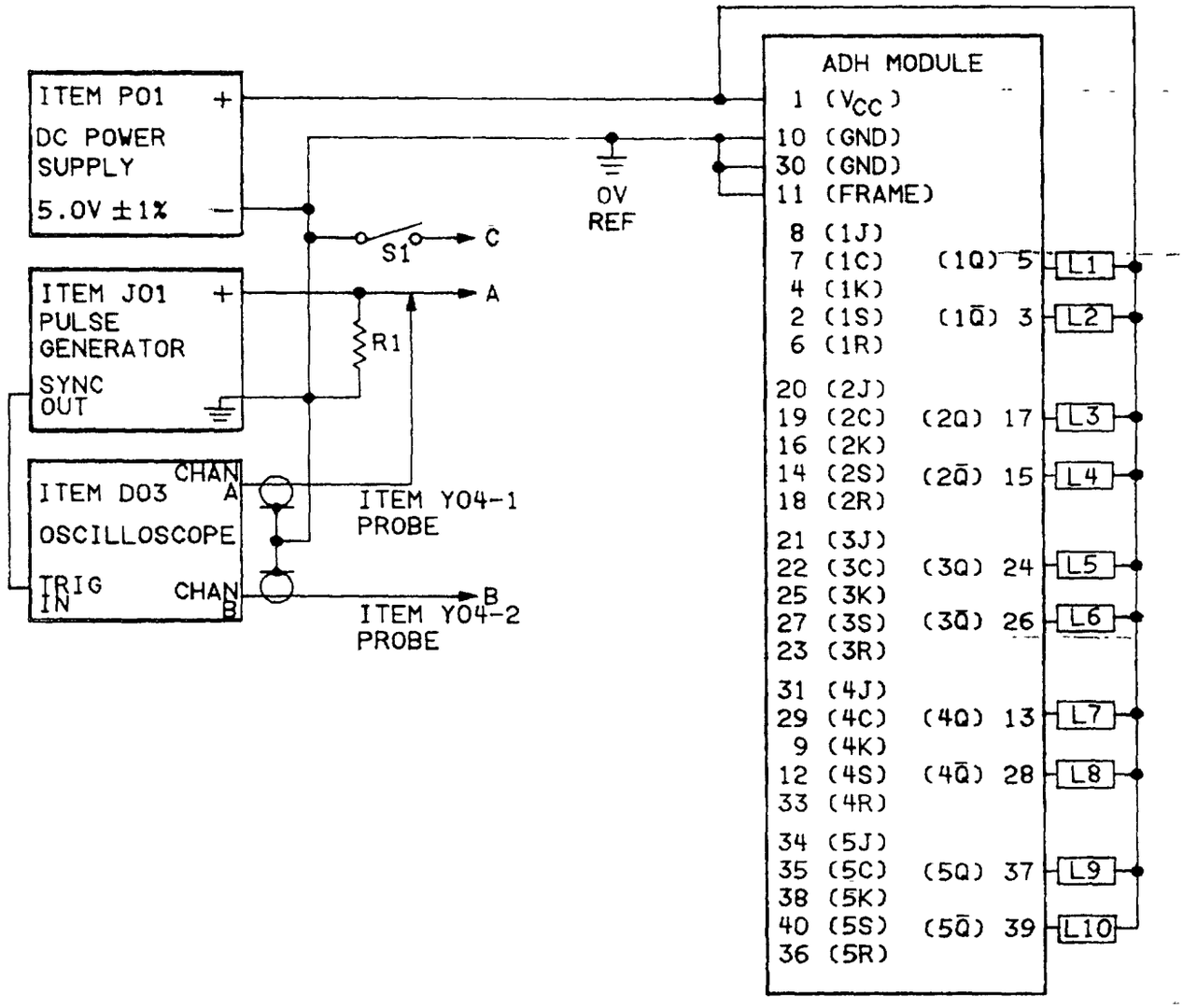


FIGURE 25. Clock threshold (high-level slow transition times and low-level) test circuit for table XVIII.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. S1 = SPST switch.
3. E1-E10 = Use load circuit 4 as shown on figure 15.
4. For test conditions EE1-EE5 and FF1-FF10 the threshold level voltage shall be adjusted using item D03 with item Y04 probes. The adjustment shall have a 5 mV/mm resolution and an accuracy of ±45 mV

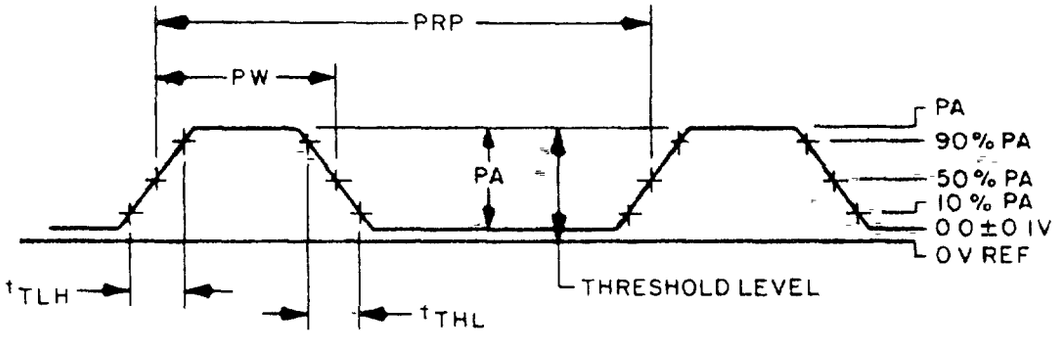
Test condition	Temperature (°C)	Threshold level (V)
EE1-EE5	0	2.0
	25	1 9
	60	1.8
FF1-FF10	0	1.2
	25	1 1
	60	0.95

0° and 60°C refer to operating temperatures and 25°C refers to ambient temperature.

5. For test conditions EE1 through EE5, an output pulse shall be present and the period of the pulse, measured at 1.0 V ±5% from the low level shall conform to the requirements of table I. The measurement shall be made using item Y04-2 and item D03
6. After making all circuit connections for each test condition (FF1 through FF10) close switch S1 for at least 100 ns, then open it and measure the output voltage with item Y04-2 and item D03

FIGURE 25. Clock threshold (high-level slow transition times and low-level) test circuit for table XVIII - Continued.

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NOTES
1 Input pulse parameters
PW = 1.25 ms ±10%
PRP = 2.5 ms ±10%
$t_{TLH} = t_{THL} = 0.5$ ms ±10%
2 Adjust the item J01 to obtain the above waveform at point A (figure 25) with A open. After A has been connected, some item J01 readjustment may be necessary due to generator loading

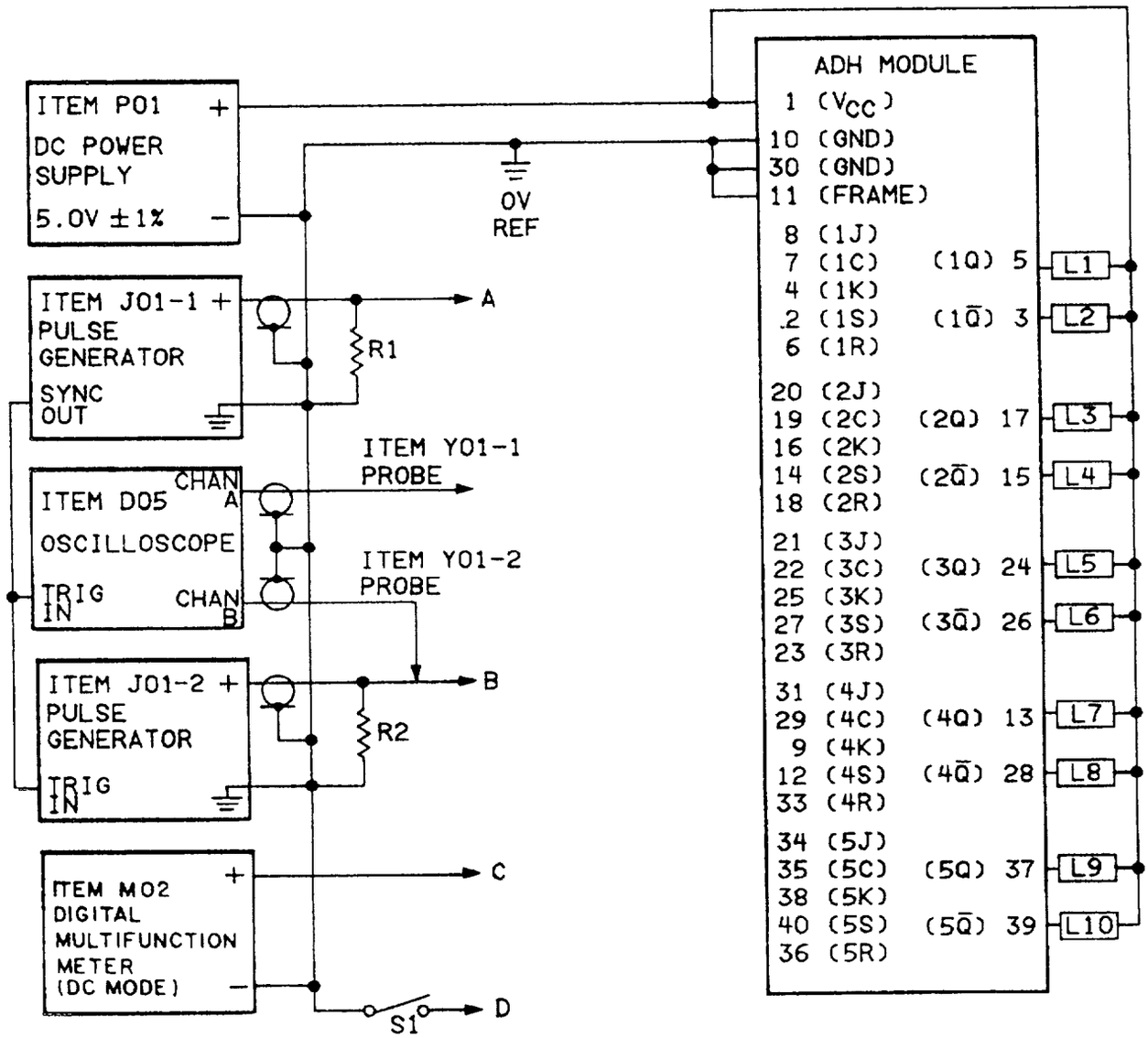
FIGURE 26. Clock threshold (high-level slow transition times and low-level) test circuit waveform for table XVIII.

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TABLE XIX Transient immunity (J and K) and minimum pulse sensitivity (J-K) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
GG1	1K Transient immunity, J and K	7	4	3	2	0.00	0.40	V	27,28
GG2	2K	19	16	15	14				
GG3	3K	22	25	26	27				
GG4	4K	29	9	28	12				
GG5	5K	35	38	39	40				
GG6	1J	7	8	5	6				
GG7	2J	19	20	17	18				
GG8	3J	22	21	24	23				
GG9	4J	29	31	13	33				
GG10	5J	35	34	37	36				
HH1	1J-1K Minimum pulse sensitivity, J-K	7	8,4	Open	Open	1.6	2.4	µs	27,29
HH2	2J-2K	19	20,16						
HH3	3J-3K	22	21,25						
HH4	4J-4K	29	31, 9						
HH5	5J-5K	35	34,38						

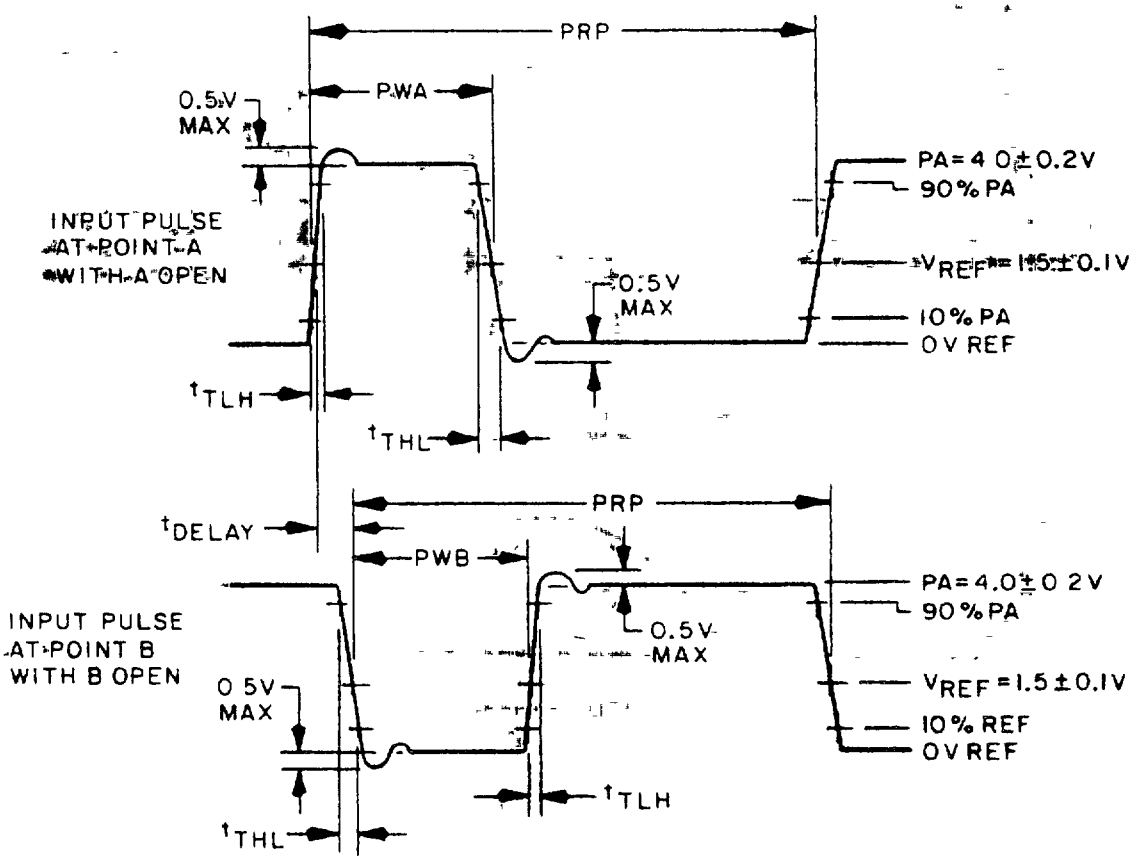
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NOTES
1. R1 = R2 = 50Ω ±5%, 1 W.
2. L1-L10 = Use load circuit 4 as shown on figure 15.
3. S1 = SPST switch.
4. Connect A and B to the appropriate module pins and readjust input voltage levels, pulse widths and delays if necessary. (Verify the 4.0 V pulse amplitude level at B using item D03 with item Y04 probe, the adjustment shall have a 5 mV/mm resolution)
5. For test conditions GG1-GG10, close switch S1 for at least 100 ns, open it and make output voltage measurement using item V03

FIGURE 27. Transient immunity (J and K) and minimum pulse sensitivity (J-K) test circuit for table XIX.

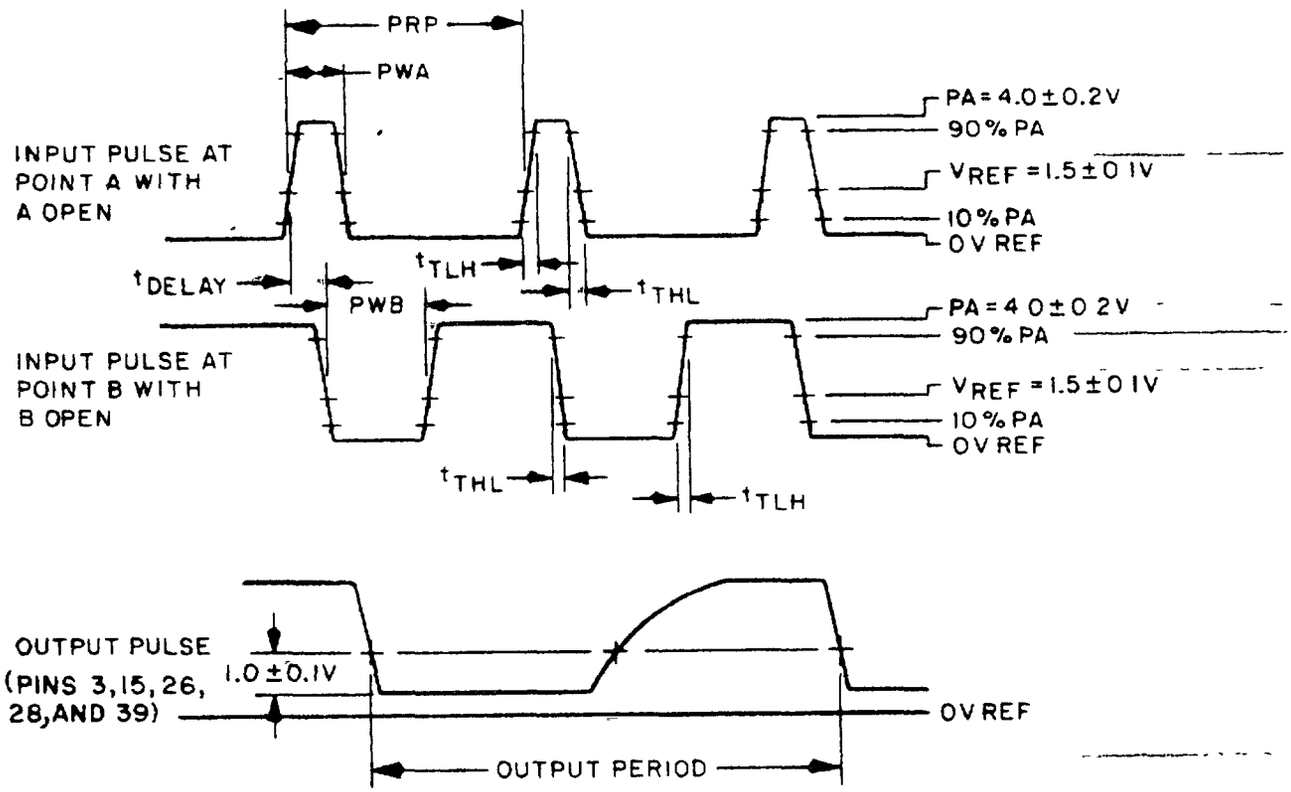
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NOTES:
1. For test conditions GG1-GG10 adjust items J01-1 and J01-2 to obtain the above input waveforms at A and B. After A and B have been connected, some pulse generator readjustment may be necessary due to generator loading.
2. Input pulse parameters
   PWA = 100 ±10 ns.
   PWB = 200 ±20 ns
   PRP = 1.0 ±0.1 µs
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   $t_{DELAY} = 8 \pm 1$ ns.

FIGURE 28. Transient immunity (J and K) test circuit waveforms for table XIX.

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NOTES
1. For test conditions HH1-HH5 adjust items J01-1 and J01-2 to obtain the above input waveforms at A and B. After A and B have been connected, some pulse generator readjustment may be necessary due to generator loading.
2. Input pulse parameters
   PWA = 100 ±10 ns
   PWB = 200 ±20 ns
   PRP = 1.0 ±0.1 us
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns
   $t_{DELAY} = 65 +5$ ns
   Overshoot and undershoot shall not exceed 0.5 V

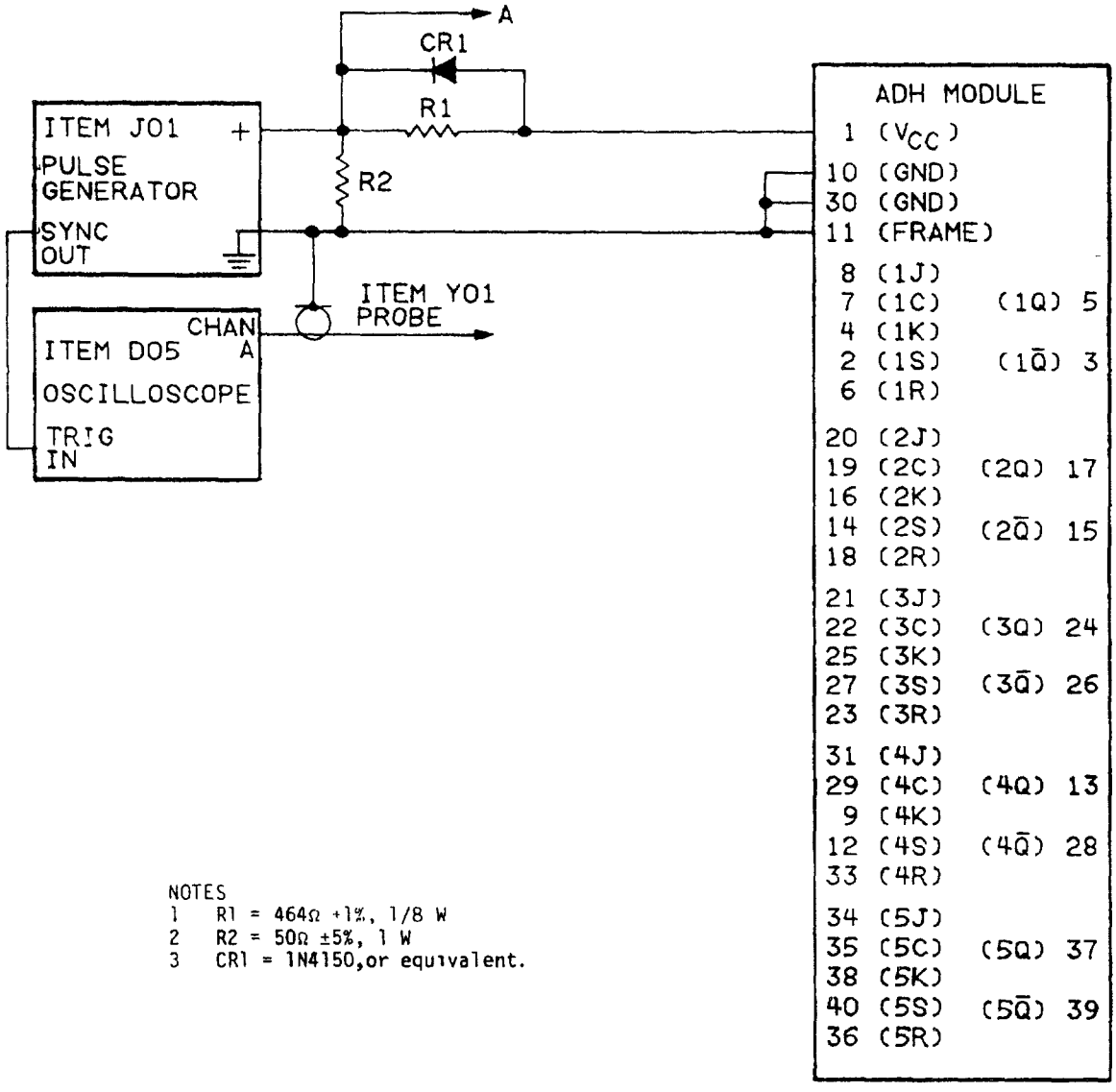
FIGURE 29. Minimum pulse sensitivity (J-K) test circuit waveforms for table XIX.

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TABLE XX. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max.		
JJ	Power supply transient rise time	36	148	µs	30,31

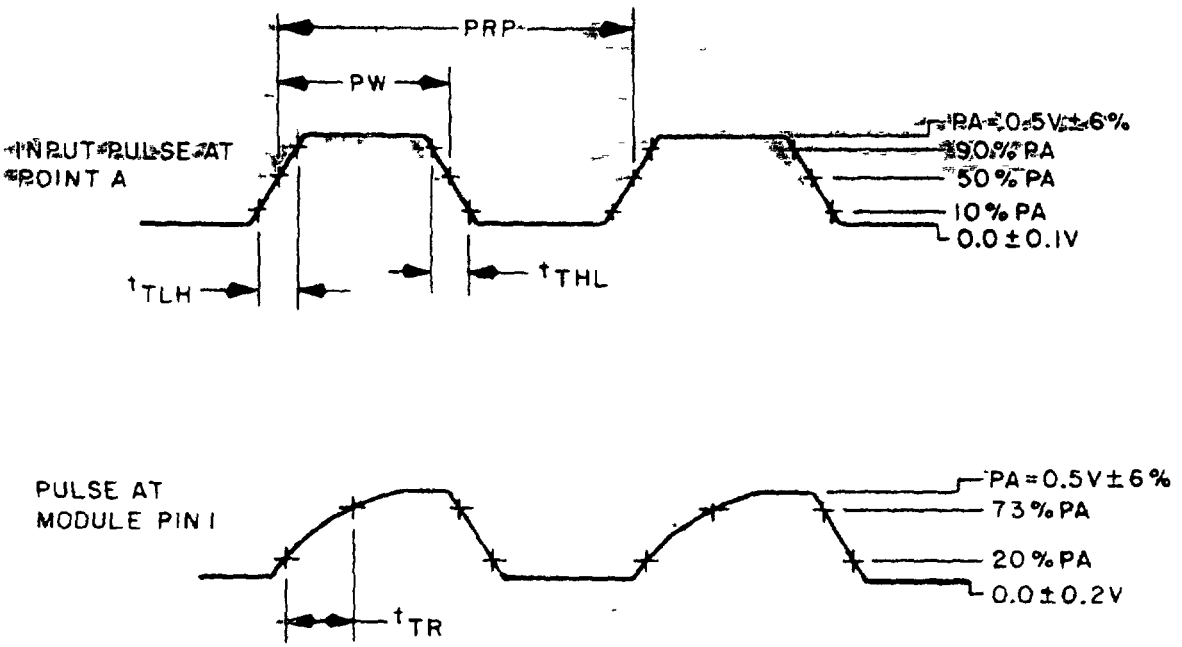
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NOTES
1 R1 = 464Ω ±1%, 1/8 W
2 R2 = 50Ω ±5%, 1 W
3 CR1 = 1N4150,or equivalent.

FIGURE 30. Power supply transient rise time test circuit for table XX.

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NOTES
1 Input pulse parameters at point A
PW = 750 µs ±5%
PRP = 1.5 ms +5%
$t_{TLH} = t_{THL}$ = 50 ns ±10%.
2 Measure t_{TR} at module pin 1

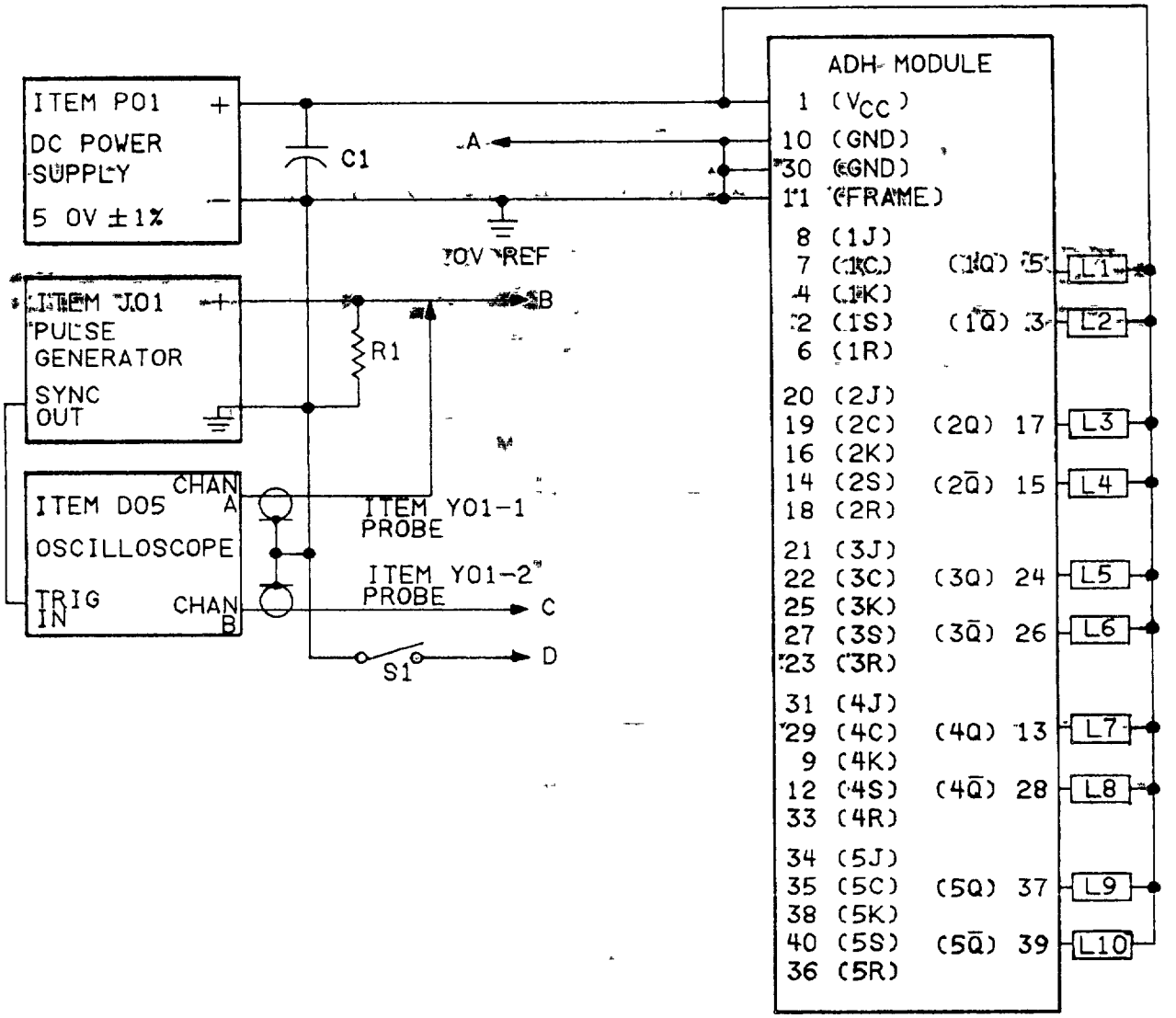
FIGURE 31. Power supply transient rise time test circuit waveforms for table XX.

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TABLE XXI. Output crosstalk (low-level and high-level) test

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
KK1	1Q Output crosstalk, low level	19,22, 29,35	14,27 12,40	6	5	0 0	0 3	V	32,33
KK2	1$\overline{Q}$ Output crosstalk, high level	19,22, 29,35	14,27 12,40	6	3				
KK3	2Q Output crosstalk, low level	7,22, 29,35	2,27 12,40	18	17				
KK4	2$\overline{Q}$ Output crosstalk, high level	7,22, 29,35	2,27 12,40	18	15				
KK5	3Q Output crosstalk, low level	7,19, 29,35	2,14 12,40	23	24				
KK6	3$\overline{Q}$ Output crosstalk, high level	7,19, 29,35	2,14 12,40	23	26				
KK7	4Q Output crosstalk, low level	7,19, 22,35	2,14 27,40	33	13				
KK8	4$\overline{Q}$ Output crosstalk, high level	7,19, 22,35	2,14 27,40	33	28				
KK9	5Q Output crosstalk, low level	7,19, 22,29	2,14 27,12	36	37				
KK10	5$\overline{Q}$ Output crosstalk, high level	7,19, 22,29	2,14 27,12	36	39				

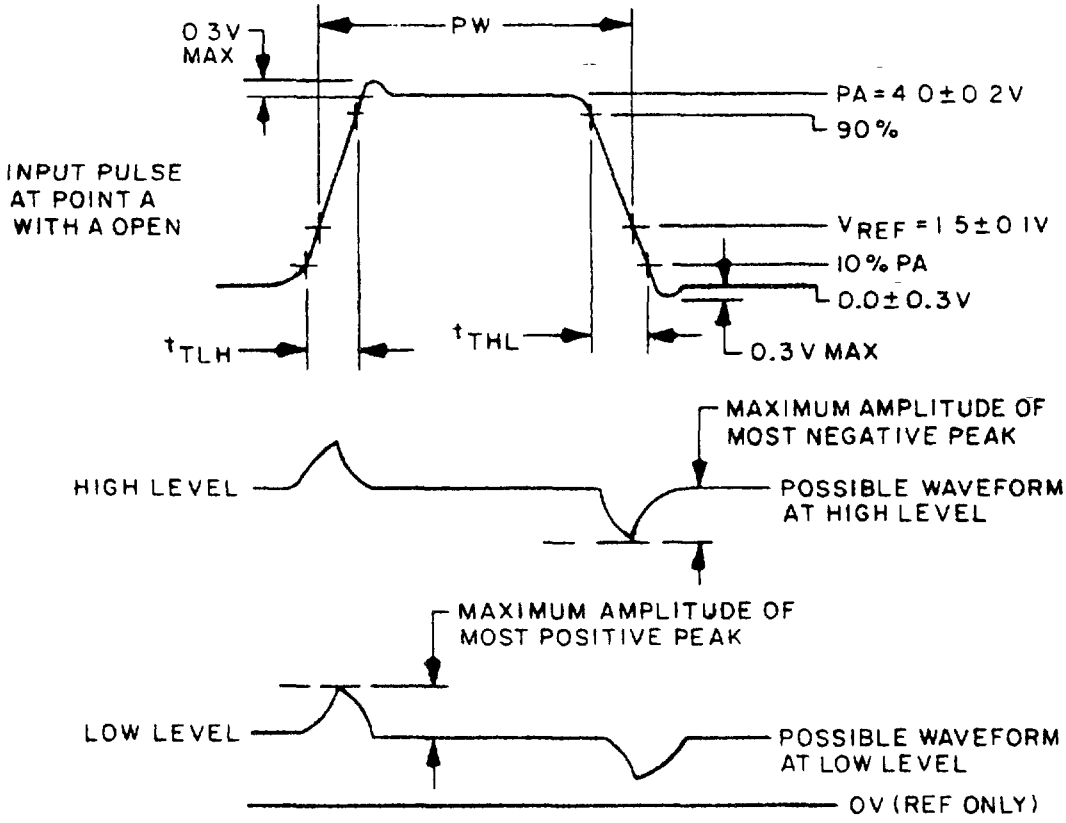
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NOTES
1. R1 = 50Ω ±5%, 1 W
2. L1-L10 = Use load circuit 3 as shown on figure 15.
3. S1 = SPST switch
4. C1 = Represents power supply bypass capacitance added to the test fixture to limit the transient response of the power supply and the power supply transmission paths to module current demands. Capacitance may be added as required.
5. After making all circuit connections for a test condition, close switch S1 for at least 100 ns and then open and make output measurement

FIGURE 32. Output crosstalk (low-level and high-level) test circuit for table XXI.

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NOTE Input pulse parameters
PW = 100 ±10 ns
PRR = 1 MHz ±10%
$t_{TLH} = t_{THL} = 20 \pm 2$ ns.

FIGURE 33 Output crosstalk (low-level and high-level) test circuit wavefroms for table XXI.

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TABLE XXII. Input capacitance test.

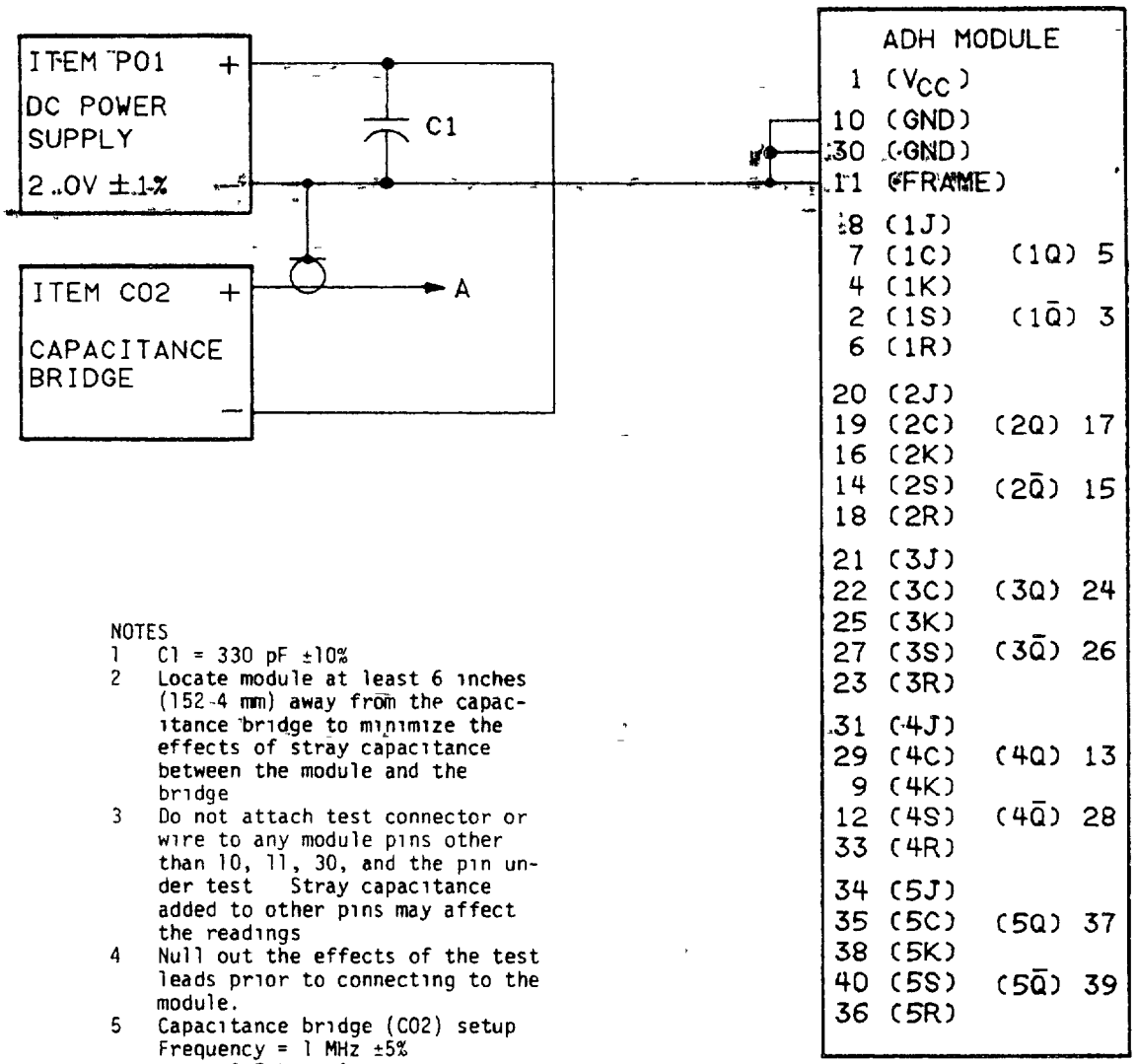
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
LL1	1S Input capacitance	2	2	16	pF	34
LL2	1J	8				
LL3	1C	7				
LL4	1K	4				
LL5	1R	6				
LL6	2S	14				
LL7	2J	20				
LL8	2C	19				
LL9	2K	16				
LL10	2R	18				
LL11	3S	27				
LL12	3J	21				
LL13	3C	22				

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TABLE XXII. Input capacitance test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
LL14	3K Input capacitance	25	2	16	pF	34
LL15	3R	23	↓	↓	↓	↓
LL16	4S	12				
LL17	4J	31				
LL18	4C	29				
LL19	4K	9				
LL20	4R	33				
LL21	5S	40				
LL22	5J	34				
LL23	5C	35				
LL24	5K	38				
LL25	5R	36				

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NOTES
1. C1 = 330 pF ±10%
2. Locate module at least 6 inches (152.4 mm) away from the capacitance bridge to minimize the effects of stray capacitance between the module and the bridge
3. Do not attach test connector or wire to any module pins other than 10, 11, 30, and the pin under test. Stray capacitance added to other pins may affect the readings
4. Null out the effects of the test leads prior to connecting to the module.
5. Capacitance bridge (CO2) setup
   Frequency = 1 MHz ±5%
   V_{pk} = 0.5 V maximum

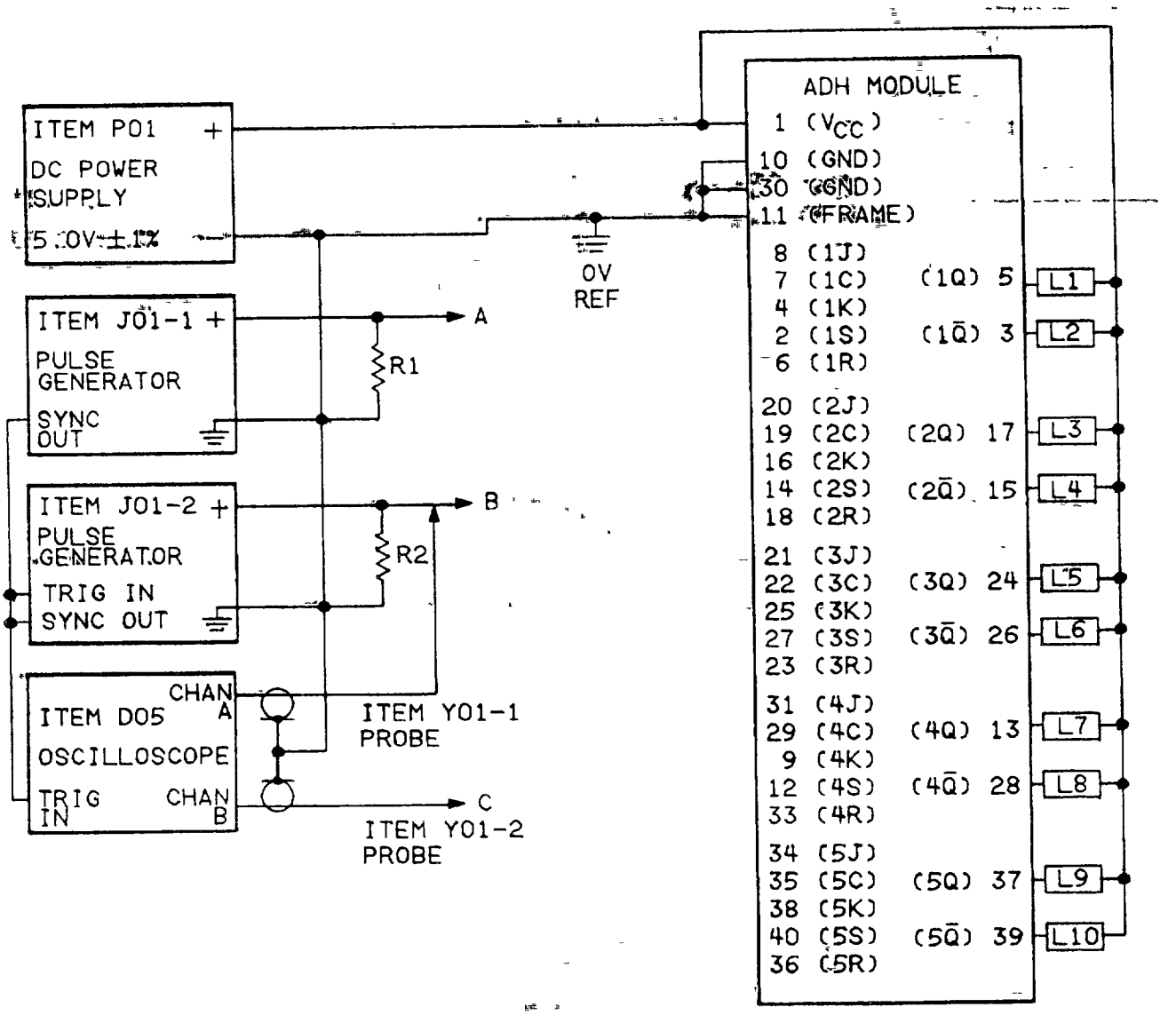
FIGURE 34. Input capacitance test circuit for table XXII.

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TABLE XXIII. Propagation delay time (t_{PHL} and t_{PLH}, R and S)test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
MM1	1R Propagation delay time, t_{PHL}, R and S	7, 8	6	5	15	80	ns	35,36
MM2	2R	19,20	18	17				
MM3	3R	22,21	23	24				
MM4	4R	29,31	33	13				
MM5	5R	35,34	36	37				
MM6	1S	7, 4	2	3				
MM7	2S	19,16	14	15				
MM8	3S	22,25	27	26				
MM9	4S	29, 9	12	28				
MM10	5S	35,38	40	39				
MM11	1R Propagation delay time, t_{PLH}, R and S	7, 8	6	3		95		35,37
MM12	2R	19,20	18	15				
MM13	3R	22,21	23	26				
MM14	4R	29,31	33	28				
MM15	5R	35,34	36	39				
MM16	1S	7, 4	2	5				
MM17	2S	19,16	14	17				
MM18	3S	22,25	27	24				
MM19	4S	29, 9	12	13				
MM20	5S	35,38	40	37				

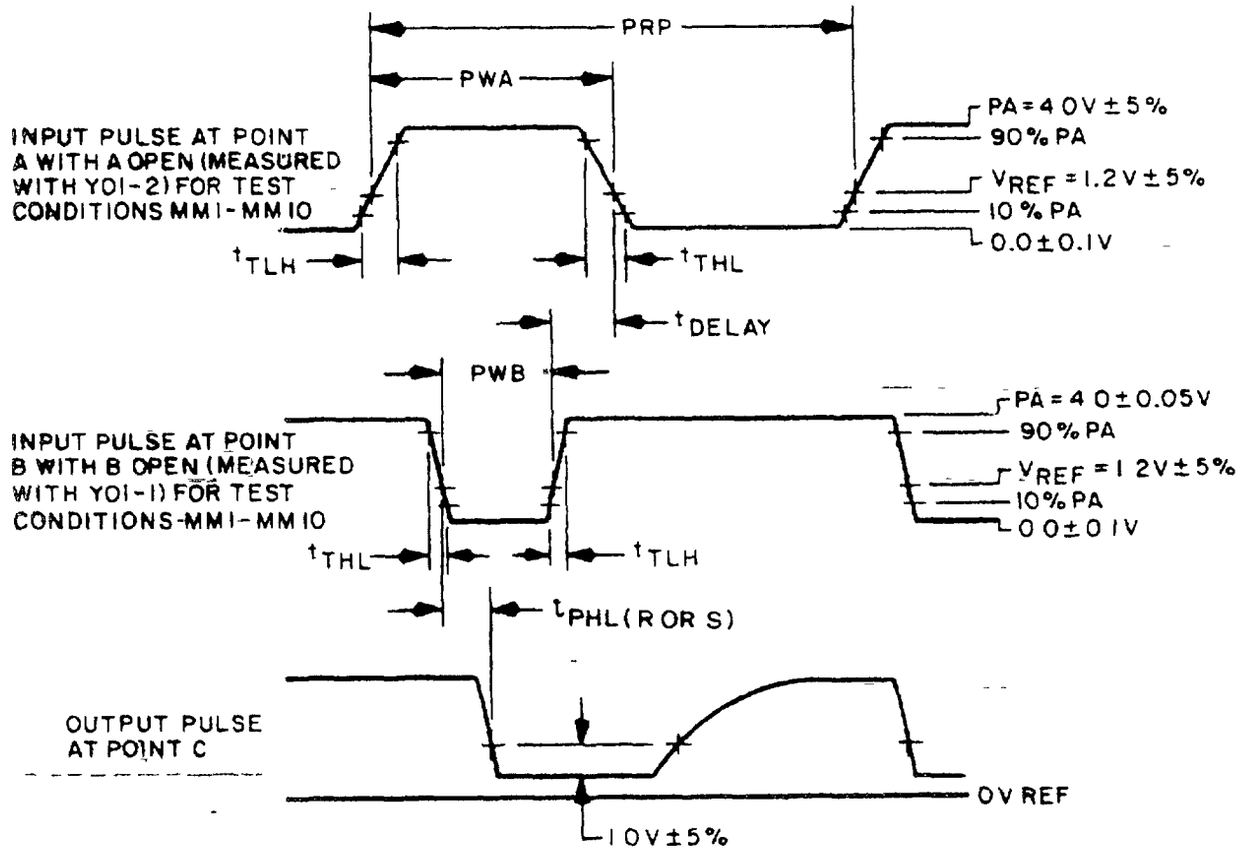
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NOTES
1 R1 = R2 = 50Ω ±5%, 1 W.
2 L1-L10 = Use load circuit 2 as shown on figure 15.

FIGURE 35 Propagation delay time (t_{PHL} and t_{PLH}, R and S) test circuit for table XXIII.

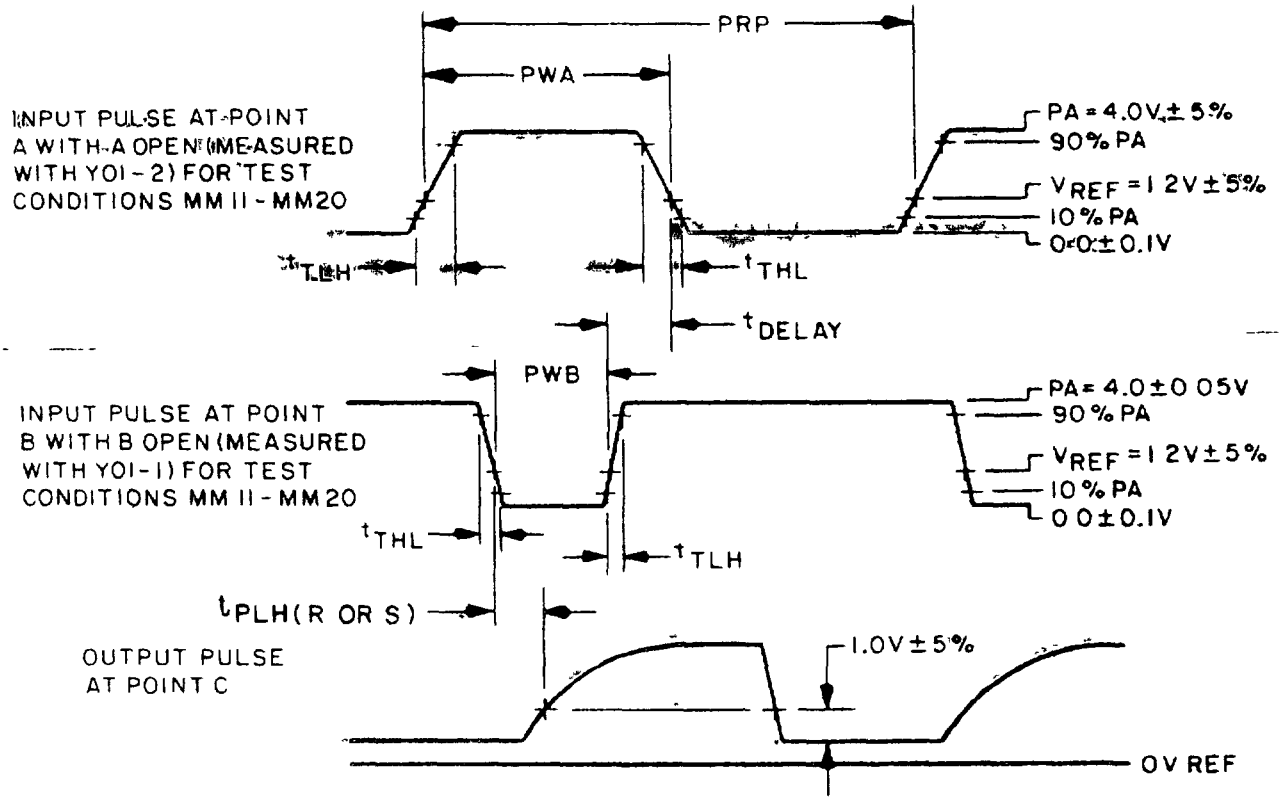
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NOTES
1. Input pulse parameters at A and B
   PWA = 600 ns ±10%
   PWB = 200 ns ±10%
   PRP = 3 µs ±10%
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
   $t_{DELAY} = 20 \pm 1$ ns
2. Input pulse shall be as above at points A and B (The 4.0 V level at B must be adjusted by using items D03 and Y04 probe The adjustment must have a 5 mV/mm resolution)
3. After input pulses at A and B have been adjusted, connect A and B per table XXIII, move item Y01-2 to the appropriate output pin and measure the turn-on delay time from the input at point B to the output

FIGURE 36. Propagation delay time (t_{PHL}, R and S) test circuit waveforms for table XXIII.

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NOTES
1. Input pulse parameters at A and B
   PWA = 600 ns ±10%
   PWB = 200 ns ±10%
   PRP = 3 µs ±10%
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns
   $t_{DELAY} = 20 \pm 1$ ns
2. Input pulse shall be as above at A and B (The 4.0 V level at B must be adjusted by using items D03 and Y04 probe. The adjustment must have a 5 mV/mm resolution.)
3. After input pulse at A and B have been adjusted, connected A and B per table XXIII, move item Y01-2 to the appropriate output pin and measure the turn-off delay time from the input at B to the output.

FIGURE 37. Propagation delay time (t_{PLH}, R and S) test circuit wavefroms for table XXIII.

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